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Trial-by-trial fluctuations in decision criterion shape confidence

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Abstract

Many decisions are accompanied by confidence. One influential view couched within Signal Detection Theory (SDT) views confidence as the distance between a decision variable and a static decision criterion. However, recent work suggests that the criterion undergoes trial-by-trial fluctuations. Combining theory and model simulations, we predict that fluctuations in the decision criterion shape confidence. In 15 datasets, trial-by-trial estimates of decision criterion were obtained with the Hierarchical Model for Fluctuations in Criterion (hMFC). Across datasets, we confirmed our pre-registered hypothesis that confidence is shaped by single-trial criterion state. This effect was found in 14 out of 15 datasets, indicating a robust pattern across a variety of paradigms and confidence scales. Beyond self-report, the effect was also observed in RTs, pupil-linked arousal and a neural signature of confidence. Our results demonstrate that variability in confidence, often treated as noise, actually reflects genuine sensitivity to the current state of the (fluctuating) decision criterion.

Introduction

Every day we have to make a multitude of decisions. With each decision comes a subjective feeling of decision confidence, indicating the probability of having made the correct choice. Many computational models have helped to understand the mechanisms that give rise to people's reported decision confidence when they make choices. Signal detection theory (SDT) is a classical and influential computational framework of decision making ¹, and easily extends to modelling decision confidence.

In its simplest form, SDT describes how observers classify noisy stimuli as belonging to one of two categories. The theory assumes that observers generate an internal representation of the relevant stimulus information, typically referred to as the decision variable. To make a binary decision, this decision variable is compared against an internal decision criterion (Figure 1; static criterion, gray line). Because the brain is inherently noisy, repeated presentations of the same stimulus do not always result in the same internal representation. Instead, a distribution of decision variables is generated (Figure 1; two distributions, one for each stimulus). This inherent variability accounts for why participants can give different responses when presented with the same stimulus multiple times ¹⁻³.

Going beyond binary decisions, SDT also explains how observers experience graded levels of confidence in the accuracy of their decisions. Specifically, decision confidence can be computed as the absolute distance between the decision variable and the decision criterion, with a larger distance indicating higher confidence ⁴⁻⁸. This implementation is referred to as the distance-to-criterion hypothesis of confidence. Although this approach can be traced back already to the early proposals of SDT ¹, in recent years this approach has been popularized within the perceptual metacognition literature.

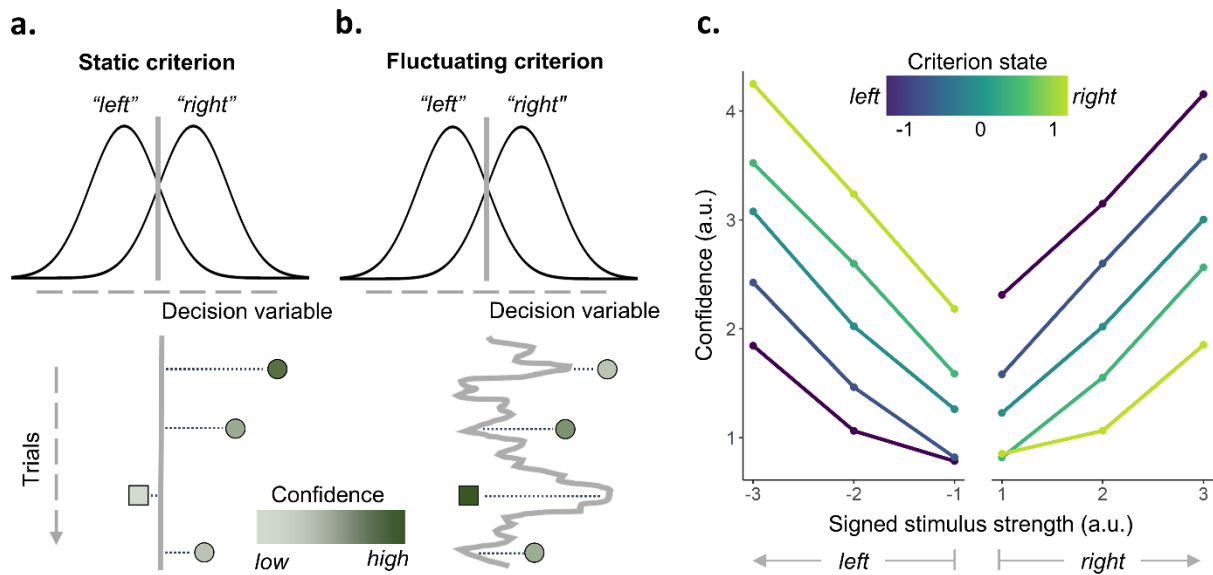


Figure 1: Simulating confidence as distance-to-criterion under trial-to-trial criterion fluctuations.

A. According to signal detection theory (SDT), choices are formed by comparing a decision variable to a criterion. An observer will respond “left” (squares) or “right” (circles), depending on whether the decision variable falls left or right to this criterion (grey line), respectively. Within SDT, confidence can be quantified as the absolute distance between the decision variable and the criterion (dotted lines), also known as the distance-to-criterion hypothesis. **B.** Whereas this criterion is typically assumed to be fixed across a series of trials (static criterion), here we investigate the consequences for decision confidence when this criterion slowly fluctuates across time. As illustrated, for identical decision variables, the predicted confidence can differ, depending on whether the criterion is static or fluctuating. **C.** Formal model simulations (6000 trials) from an SDT framework with a fluctuating criterion (i.e., as illustrated in Figure 1B), show a clear influence of criterion state on confidence. For left stimuli (negative values on x-axis), confidence is higher when the criterion fluctuates towards the right (lighter colors). This pattern flips for right stimuli (positive values on x-axis), with higher confidence when the criterion fluctuates towards left (darker colors). Note: a.u. = arbitrary unit.

A key component in signal detection theory is the decision criterion to account for biases (i.e., when one response option is chosen more often than the other). For example, shifting the criterion to the right will result in overall more ‘left’ responses, while leaving the overall sensitivity unaffected. A large body of empirical work shows that human observers strategically shift the decision criterion in response to external feedback⁹, the base-rate of stimuli¹⁰, task instructions¹¹, expectancy¹², previous choices^{13,14}, and monetary rewards¹⁵. To assess whether an observer uses a shifted (or biased) decision criterion, its position is calculated based on the hit and false alarm rate. Crucially, by doing so it is implicitly assumed that the criterion is fixed, and thus that this (potentially shifted) criterion is the same for all trials (Figure 1A; static criterion). However, increasing evidence suggests that computational

variables, like the decision criterion, are not static but instead fluctuate at the level of individual trials (Figure 1B; fluctuating criterion) ^{16–22}.

In the signal detection theory framework, trial-by-trial criterion fluctuations should also affect the computation of confidence, given that confidence is thought to reflect the distance between the criterion and the decision variable (Figure 1B). More specifically, when the criterion fluctuates towards the right, on average the distance between a decision variable favoring the “left” stimulus and the criterion will increase, leading to higher confidence. On the contrary, when the criterion fluctuates towards the left the distance between a decision variable favoring the “left” stimulus and the criterion will decrease, leading to lower confidence. Indeed, exactly this pattern is observed when formally simulating data from an SDT observer with a fluctuating criterion (Figure 1C, see Methods and Results for detailed description). These simulations also show that given the same stimulus, confidence can vary quite substantially. Thus, this suggests that variability in confidence reports, which have traditionally been treated as noise ⁶, might actually reflect genuine sensitivity of confidence judgments to the current state of the (fluctuating) decision criterion.

Here, we build on current developments in computational modeling which allow to effectively measure trial-to-trial fluctuations in the decision criterion. Vloeberghs and colleagues (2025) developed the hierarchical Model of Fluctuations in Criterion (hMFC) ²⁰, which uses a hierarchical Bayesian estimation approach to accurately recover criterion estimates at the single trial level (see Methods for detailed description). This new modelling framework now makes it possible to put the hypothesis to the test.

In sum, the current study aims to examine whether self-reported confidence ratings, reaction times, and neurophysiological measures of confidence are influenced by trial-to-trial fluctuations in the decision criterion, quantified with hMFC. We first report results from a pre-registered analysis of 15 behavioral datasets, which confirmed the influence of criterion fluctuations on confidence ratings. Analyzing these datasets individually, the effect was found in 14 out of 15 datasets. In two additional datasets, these findings were replicated using an implicit measure of confidence, namely reaction times, and two neurophysiological markers of confidence, namely pupil size and a neural EEG signature of confidence. Together, these results suggest a robust effect that is present in a variety of paradigms and confidence scales and show that variability in both behavioral and neurophysiological measures of confidence is driven by fluctuations in the underlying latent decision criterion.

Results

Model simulations

To investigate whether and how confidence is shaped by criterion fluctuations we first simulated data from an SDT observer with the criterion varying from trial to trial according to a first-order autoregressive process (AR1) (Figure 1C). First, confidence increases as stimulus strength increases ($\beta = 0.840$, $SE = 0.014$, $F(1) = 3313.823$, $p < .001$, $CI(95) = [0.811, 0.868]$). More interestingly, the interaction between criterion and stimulus direction, which is central to our hypothesis, is significant ($\beta = -0.423$, $SE = 0.035$, $F(1) = 3674.507$, $p < .001$, $CI(95) = [-0.492, -0.355]$). Specifically, we find that confidence decreases for a “right” stimulus when the criterion fluctuates to the right and increases when it fluctuates to the left. For a “left” stimulus, the opposite pattern emerges. In addition, we observe a significant three-way interaction between criterion, stimulus direction, and stimulus strength ($\beta = -0.191$, $SE = 0.016$, $F(1) = 138.295$, $p < .001$, $CI(95) = [-0.223, -0.159]$). Note that the significance of the three-way interaction is not a reliable signature for confidence as distance-to-criterion. Instead, it arises from a specific combination of stimulus configuration and the range of criterion fluctuations. Specifically, the three-way interaction will be significant when the lines in Figure 1C are not parallel, which occurs when the dataset contains trials in which the criterion coincides with the mean of the generative SDT stimulus distribution (e.g., criterion of -1 with a stimulus of -1, or a criterion of +1 with a stimulus of +1). In Supplementary Figure 1, we simulated an agent with a smaller range of the criterion fluctuations where such trials did not occur. For this agent, although the critical two-way interaction between stimulus direction and criterion is significant, the three-way interaction is not, providing evidence that this is not a key model prediction.

To assess how the results change as a function of accuracy we performed separate analyses for correct and error trials (Suppl. Figure 2). The pattern for correct trials mirrors that of the full dataset, whereas for error trials the interaction between stimulus direction and criterion reverses, with higher confidence for leftward stimuli when the criterion fluctuates leftward.

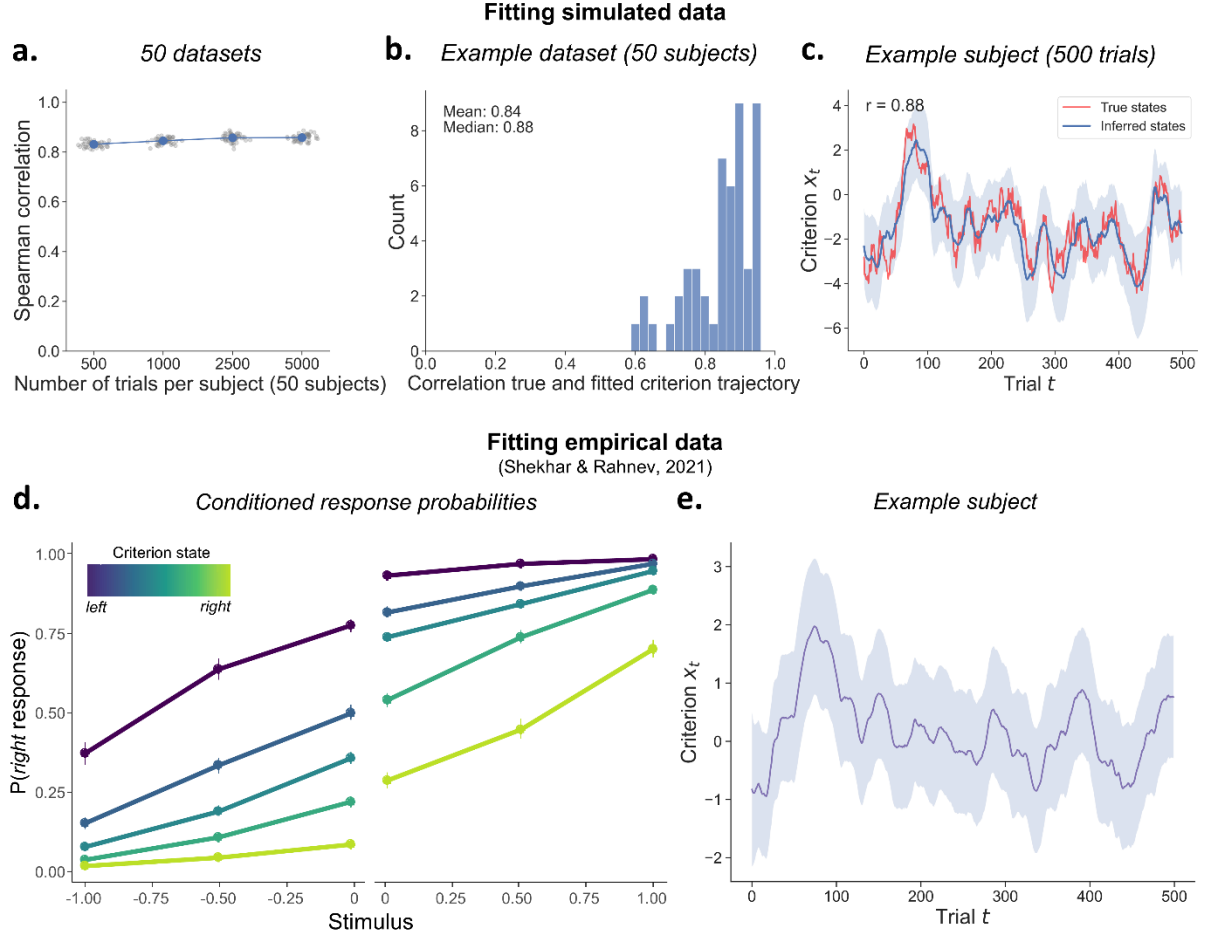


Figure 2: Estimating criterion fluctuations with hMFC on simulated and empirical data. **A.** The recoverability of the criterion fluctuations by hMFC was assessed in Vloeberghs et al. (2025)²⁰ using simulated data with a varying number of trials per subject (500, 1000, 2500 or 5000 trials), with 50 subjects per dataset and 50 datasets per trial count. Each dot represents the correlation between the true and estimated criterion trajectory averaged over subjects in one dataset. The error bars are plotted but are very small. **B.** The correlation between true and estimated criterion trajectory for each subject (with 500 trials) in a representative simulated dataset. **C.** The true and estimated criterion fluctuations for a simulated subject with 500 trials. The line for the estimated fluctuations shows the posterior mean and the shaded area indicates the 95% credible interval of the posterior at each trial. **D.** The empirical response probabilities of the data (20 subjects with 2800 trials each) of Shekhar and Rahnev (2021, Experiment 4)⁶ conditioned on the binned trial-by-trial criterion. The more the criterion is shifted to the left, the more right responses are observed. Error bars reflect standard errors. **E.** The estimated criterion trajectory of an example subject (first 500 trials are shown). The line indicates the posterior mean of each trial with the shaded area representing the 95% credible intervals.

Empirical data

Following these simulations, we next investigated our hypothesis in empirical data, estimating criterion fluctuations using hMFC ²⁰. hMFC can accurately recover the criterion fluctuations in simulated data, even with only 500 trials per subject (Figure 2A-C). In empirical data, the trial-by-trial criterion estimated by hMFC captures differences in response probabilities that are consistent with criterion shifts in SDT, as illustrated by the response probabilities conditioned on criterion state shown in Figure 2D. The more the criterion fluctuates to the right, the more left responses are observed. Conversely, a leftward shift in the criterion leads to more rightward responses. The model fits show stable and interpretable posterior distributions, with proper mixing of the chains of the Gibbs sampler (all R-hat < 1.05), reassuring that we sampled from the full posterior range (Suppl. Figure 3 and Suppl. Table 1). Supplementary Table 2 reports the mean, standard deviation, minimum, and maximum values, calculated per study, for the per-subject parameters of the autoregressive model that characterizes the criterion fluctuations. All together, these results ensure the validity of the hMFC estimates.

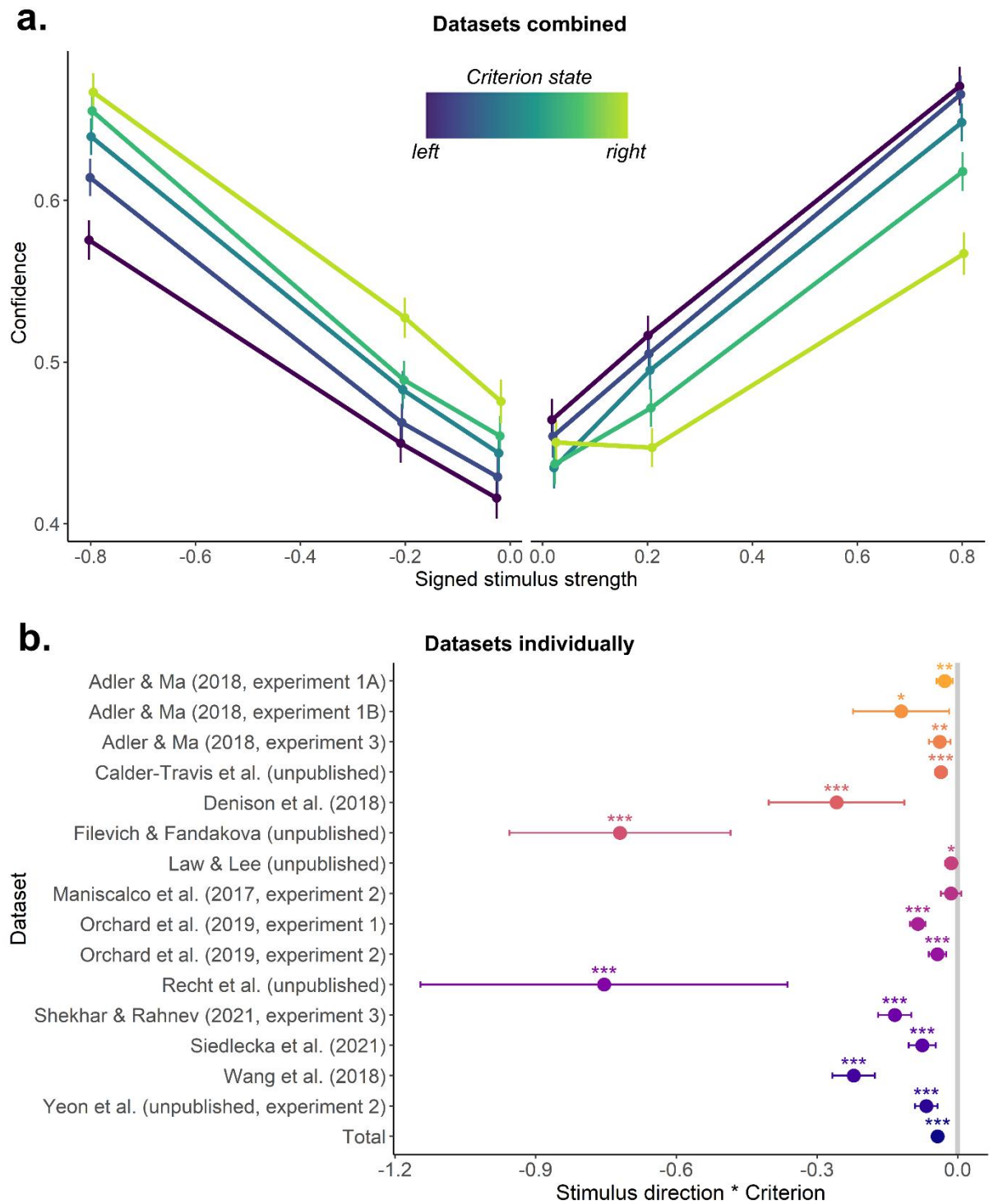


Figure 3: Trial-by-trial fluctuations in decision criterion shape confidence ratings in empirical data. **A.** As theoretically predicted, and in line with model simulations, decision confidence was modulated by trial-by-trial fluctuations in decision criterion. Specifically, when the stimulus direction was ‘right’ (positive signed stimulus strength) confidence increases as the decision criterion fluctuates towards the left (i.e. becomes more blue) and confidence decreases as the decision criterion fluctuates towards the right (i.e., become more yellow). Data are shown across the 15 datasets. Error bars reflect standard errors. For plotting purposes only, criterion states were binned and the average within each bin is shown. **B.** Parameter estimates for the crucial interaction between stimulus direction and decision

*criterion and their 95% confidence intervals. The estimates are obtained from linear mixed models fitted separately to every dataset. The estimate plotted at the bottom, labelled “Total”, shows the parameter estimate from the model fitted to all datasets (taking into account the nesting of subjects in experiments). The number of subjects and trials in each dataset are reported in Supplementary Table 4. The p-values were obtained using two-sided tests. Note: * = $p < .05$, ** = $p < .01$, *** = $p < .001$.*

Next, we turn towards the key question of the current work: is confidence shaped by trial-to-trial fluctuations in decision criterion? In empirical data we observed the same confidence patterns as seen in the simulations (Figure 3A). Indeed, when fitting empirical data using a linear mixed model in which confidence is predicted based on stimulus strength, stimulus direction, and criterion (Model 1; see Methods) we found a highly significant interaction between stimulus direction and criterion, hereby confirming our main hypothesis ($\beta = -0.043$, $SE = 0.002$, $F(1,456507) = 781.961$, $p < .001$, $CI(95) = [-0.046, -0.040]$). The negative slope indicates lower confidence for a right stimulus (+1) when the criterion drifts towards the right (positive values), and higher confidence when the criterion drifts towards the left (negative values). The opposite is true for left stimuli (-1). Additionally, the three-way interaction was significant, similar to the simulations ($\beta = -0.032$, $SE = 0.003$, $F(1,456276) = 123.330$, $p < .001$, $CI(95) = [-0.038, -0.026]$), with a larger interaction between stimulus direction and criterion for stronger stimulus strength. Next, we found a significant effect of stimulus strength, with confidence increasing as stimulus strength increases ($\beta = 0.587$, $SE = 0.114$, $F(1,14) = 26.715$, $p < .001$, $CI(95) = [0.364, 0.810]$). Lastly, the main effect of stimulus direction was borderline significant ($\beta = -0.002$, $SE = 0.001$, $F(1,456483) = 3.665$, $p = .056$, $CI(95) = [-0.004, 0.000]$). All other effects were not significant (all $p \geq .144$). No multicollinearity problems were detected (all $VIF \leq 2.607$). Through a sensitivity analysis we ensured that the results were identical for different values of the fixed hyperparameters μ_{σ^2} and β_{σ^2} (Suppl. Table 3).

When analyzing correct and error trials separately, we found a similar pattern as in the full dataset for correct trials (Suppl. Figure 4). Consistent with the simulations, for error trials we found that the interaction between stimulus direction and criterion flips, with higher confidence for left stimuli when the criterion fluctuates towards the left. The model selection results can also be found in Supplementary Figure 4.

To investigate the robustness of the critical two-way interaction between stimulus direction and criterion, we reanalyzed all datasets with confidence measured on 3-, 4-, or 6-point rating scales using an ordinal mixed regression model (Suppl. Figure 5). Of the 12 datasets suitable for this analysis, 10 showed a significant interaction. The remaining two datasets were not significant ($p = 0.061$ and $p = 0.066$).

To assess whether these findings reflect meaningful contributions of criterion fluctuations and not just statistical noise or model overfitting we turned to model comparison. Specifically, we tested whether the linear mixed model reported above (Model 1) outperformed a simpler model that lacked criterion fluctuations (Model 2). Results provide unequivocal support that the more complex model including criterion is preferred, even when controlling for complexity ($\Delta AIC = -7482$, $\Delta BIC = -7372$, $\chi^2(10) = 7502$, $p < .001$). This indicates that a significant portion of variability in decision confidence can be explained by fluctuations in the decision criterion.

To further rule out the possibility that these results reflect spurious correlations that can occur between slowly drifting variables²³, we implemented the session permutation method²⁴. In this approach, criterion fluctuations were randomly shuffled *across* participants (Model 3) and compared to the model with the original fluctuations (Model 1). Once again, the comparison strongly favored Model 1 ($\Delta AIC = -8091$, $\Delta BIC = -7848$). In Model 3, the shuffled criterion fluctuations failed to account for any variability in confidence, as there was no interaction between stimulus direction and criterion ($\beta = -0.002$, $SE = 0.002$, $F(1,137574) = 1.626$, $p = .202$, $CI(95) = [-0.006, 0.001]$). Also the three-way interaction between stimulus direction, stimulus strength, and criterion was absent ($\beta = 0.001$, $SE = 0.003$, $F(1,269873) = 0.182$, $p = .700$, $CI(95) = [-0.005, 0.008]$). Only the main effect of stimulus strength remained significant ($\beta = 0.592$, $SE = 0.116$, $F(1,14) = 26.228$, $p < .001$, $CI(95) = [0.364, 0.820]$). For full comparability, we report the model comparison for Model 2 fitted to the truncated dataset in Supplementary Analysis 1.

In the previous, we have shown that *across* 15 datasets we found a highly robust association between confidence and the single-trial state of the decision criterion. Next, we further unraveled to what extent this pattern was present in each of the individual datasets. To this end, we fitted the model to each dataset separately. Importantly, the key two-way interaction between stimulus direction and criterion, central to our hypothesis, was significant in 14 out of 15 datasets (Figure 3B). The three-way interaction between stimulus direction, stimulus strength, and criterion, which also was predicted by the simulations, was significant in 10 out of 15 datasets, with 7 showing a negative slope and 3 showing a positive slope. In the remaining 5 datasets the interaction was absent (Suppl. Figure 6). In summary, although the three-way interaction is noisier and more difficult to detect at the dataset level, the consistent presence of the crucial interaction for our main hypothesis between stimulus direction and criterion in 14 out of 15 datasets demonstrates that the effect is robust and generalizes across diverse paradigms and confidence rating scales.

Having established that criterion fluctuations shape confidence ratings, we next explored whether this is also the case for an implicit measure of confidence (RTs) and for neurophysiological correlates of confidence. While ratings provide valuable subjective reports, they rely on participants' introspection and may be influenced by biases or limitations in self-assessment. To complement this, we examined more objective indices of confidence that do not depend on introspective accuracy, namely reaction times, pupil size, and a neural marker of confidence (error positivity).

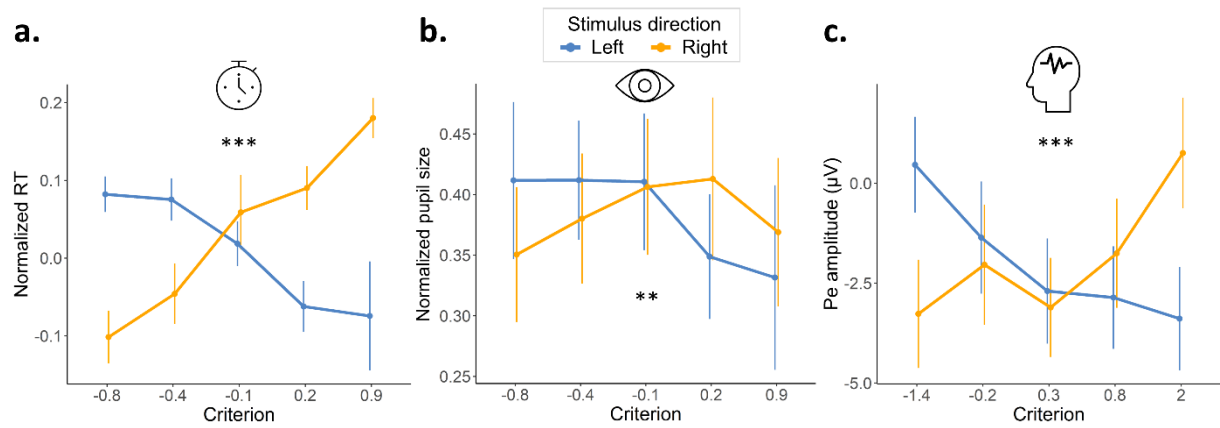


Figure 4: Criterion fluctuations shape implicit and neurophysiological measures of confidence.

Reaction times (A), pupil size (B), and a neural marker of confidence (C) are all significantly predicted by the interaction between stimulus direction and criterion in linear mixed models (reaction times: $p < .001$, pupil sizes: $p = .003$, Pe amplitude: $p < .001$). Specifically, when the criterion fluctuates to the right (positive values) for a rightward stimulus (yellow line), longer reaction times, larger pupil sizes, and a higher Pe amplitude are observed, corresponding to low confidence. In contrast, when the criterion fluctuates to the left (negative values) for a rightward stimulus, lower values for all three measures are observed, indicating high confidence. The figure shows the mean data, with error bars denoting the standard error across subjects. Reaction times and pupil size show data from Urai et al. (2017; 27 subjects with 500 trials each) ⁷, while Pe amplitude was taken from Boldt & Yeung (2015; 16 subjects with 864 trials each). In the latter dataset, only one level of stimulus strength was included, so it was not possible to show the data as a function of stimulus strength (as in Figure 3A). For consistency across panels, we therefore present only the two-way interaction. The p-values were calculated using two-sided tests. Note: * = $p < .05$, ** = $p < .01$, *** = $p < .001$. Icons in this figure are from the Microsoft Word icon library.

First, we consider reaction times. Reaction times are regarded as an implicit behavioral measure of confidence, with longer reaction times corresponding to lower confidence ^{7,25}. When predicting reaction times there was a significant effect of stimulus strength ($\beta = -0.183$, $SE = 0.014$, $F(1,26.25) = 210.460$, $p < .001$, $CI(95) = [-0.211, -0.155]$), with higher stimulus

strength (i.e., easier stimuli) resulting in lower reaction times. Crucially, the interaction between stimulus direction and criterion was significant ($\beta = 0.366$, $SE = 0.053$, $F(1,23.34) = 47.666$, $p < .001$, $CI(95) = [0.262, 0.470]$), indicating longer reaction times when the criterion fluctuates to the right for rightward stimuli (+1). Conversely, for leftward stimuli (-1), a rightward-shifted criterion resulted in shorter reaction times (Figure 4C). We further found a significant three-way interaction between stimulus direction, stimulus strength, and criterion ($\beta = 0.026$, $SE = 0.007$, $F(1,1996.98) = 13.171$, $p < .001$, $CI(95) = [0.012, 0.041]$). All other effects were not significant (all $p \geq .156$). It should be noted that this model showed some multicollinearity, with VIF scores reaching up to 10.25. Excluding the three-way interaction reduced VIFs to ≤ 3.185 , while leaving all other results unchanged.

Second, we examined a well-known neurophysiological marker of confidence: pupil size^{7,26}. Phasic arousal, as quantified through pupil size, has been shown to be inversely related to confidence, with smaller pupil size reflecting higher confidence²⁶. When predicting pupil size a significant main effect of stimulus strength emerged ($\beta = -0.045$, $SE = 0.007$, $F(1,26) = 38.757$, $p < .001$, $CI(95) = [-0.058, -0.032]$). Specifically, easier stimuli were associated with smaller pupil sizes, supporting the idea that pupil size is inversely related to confidence, since higher confidence is expected for easier stimuli. More importantly, we found that the crucial interaction between stimulus direction and criterion was significant ($\beta = 0.064$, $SE = 0.021$, $F(1,67112) = 8.992$, $p = .003$, $CI(95) = [0.022, 0.106]$). A right-shifted criterion was associated with larger pupil sizes for rightward stimuli compared to leftward stimuli, and vice versa (Figure 4A). Finally, also the three-way interaction was significant ($\beta = 0.018$, $SE = 0.008$, $F(1,67123) = 5.307$, $p = .021$), as well as the interaction between stimulus direction and stimulus strength ($\beta = 0.013$, $SE = 0.005$, $F(1,67126) = 7.292$, $p = .007$, $CI(95) = [0.003, 0.032]$). No other effects were present (all $p \geq .294$) and multicollinearity issues were absent (all $VIF \leq 3.161$).

Lastly, we investigated a neural marker of confidence, namely the error positivity measured with EEG (Pe)²⁷. The error positivity is sensitive to graded levels of confidence, with its amplitude decreasing as confidence increases^{27,28}. When predicting single-trial Pe amplitudes, we found the critical interaction between stimulus direction and criterion ($\beta = 1.912$, $SE = 0.203$, $F(1,764.50) = 89.23$, $p < .001$, $CI(95) = [1.516, 2.310]$), suggesting higher amplitudes for a rightward stimulus when the criterion fluctuates to the right, and lower amplitude when it fluctuates to the left (Figure 4B). Also the main effect of criterion was significant ($\beta = -0.476$, $SE = 0.150$, $F(1,13086.30) = 19.786$, $p < .001$, $CI(95) = [-0.770, -0.182]$). In this experiment there was only one level of stimulus strength, so the main effect of stimulus strength was omitted from the model. No multicollinearity issues were present (all $VIF \leq 1.942$). Note that the same key finding in the pre-registered analysis on 15 datasets (i.e., interaction

between stimulus direction and criterion) was also observed when analyzing the confidence ratings of this study ($\beta = -1.082$, $SE = 0.086$, $F(1, 12.95) = 156.931$, $p < .001$, $CI(95) = [-1.252, -0.913]$) with no other effects present (all $p \geq .564$). Note that the analysis performed on correct and error trials separately for all three confidence markers can be found under Supplementary Analysis 2.

For all these three markers of confidence, the model including criterion outperformed the model without them (all $\Delta AIC \leq -78$, $\Delta BIC \leq -41$), and it was also preferred over the model with shuffled criterion trajectories (all $\Delta AIC \leq -25$, $\Delta BIC \leq -61$). Together, these results demonstrate that behavioral and neurophysiological markers of confidence are shaped by fluctuations in the decision criterion.

Discussion

Within signal detection theory, decision confidence is traditionally conceptualized as the absolute distance between a decision variable and a criterion, which is known as the distance-to-criterion hypothesis. Crucially, in all previous work the criterion is assumed to be static. However, an increasing body of research suggests that computational parameters, such as the decision criterion, are dynamic and fluctuate over trials. Based on the distance-to-criterion hypothesis, we thus predict that fluctuations in decision criterion influence confidence ratings. Using the recently developed hierarchical Model for Fluctuations in Criterion (hMFC)²⁰, which allows to obtain trial-by-trial estimates of the decision criterion, we tested this hypothesis. Across fifteen datasets, we found strong and consistent evidence for this hypothesis, demonstrating a robust effect independent of the specific paradigm or confidence scale. In addition, we replicated the effect in pupil size, in a neural marker of confidence and in RTs, suggesting that criterion fluctuations constitute a source of variability in confidence across both behavioral and neurophysiological measures.

The current work provides strong evidence in favor of the distance-to-criterion hypothesis of confidence: across 17 datasets we found that seemingly random variability in decision confidence actually tracks fluctuations in decision criterion. For confidence ratings, this effect was significant in 15 out of 16 datasets (including the dataset used for the neural marker of confidence), showing that this reflects a strong and highly replicable finding. Although the interaction was not significant in one dataset, the direction of the effect was consistent with the hypothesis. From a statistical point of view, finding one null effect across 16 datasets is not surprising: given experimental power of .8 it can be expected that 3 out of 16 studies show a null effect even when the effect is real. The one study where the effect was not found was not an obvious outlier in terms of design (i.e., a Gabor detection task both with

4-choice confidence reporting) nor estimated hMFC parameters (Suppl. Table 2). Moreover, we replicated our findings across different neurophysiological measures of confidence indicating that the sensitivity to a fluctuating criterion reflects a fundamental aspect of how confidence is computed rather than merely a bias in how it is reported. In the current study we used pupil size as a neurophysiological measure of confidence. While pupil size is considered being a proxy of confidence^{7,26}, pupil size has also been linked to arousal, with fluctuations in neuromodulatory systems such as norepinephrine and acetylcholine associated with changes in pupil diameter^{29,30}. Consequently, it remains an open question to what extent some of our findings reflect fluctuations in these neurotransmitter systems.

Our findings show that a portion of variability in confidence ratings, which is typically interpreted as the results of some form of noise, can be attributed to fluctuations in the decision criterion. Models not considering the possibility of fluctuations in the criterion typically attribute variability in confidence to some form of metacognitive noise that selectively perturbs the computation of confidence^{6,31–33}. However, our current findings, unravelling the sensitivity of confidence to criterion fluctuations, advocate for a well-calibrated metacognitive system, at least with respect to criterion fluctuations. Importantly, our findings do not argue against the existence of metacognitive noise per se. It is very likely that variability in confidence is driven by fluctuations in both decision criterion, reported in the current work, and noisy confidence criteria, as put forward by research on metacognitive noise. In future work, the respective contributions of fluctuations in criterion and metacognitive noise in explaining variability in confidence could be studied through a key distinction: fluctuations in the decision criterion affect both responses and confidence ratings, whereas metacognitive noise only affects confidence ratings. While such a model does not yet exist, an alternative approach would be to determine the relative contribution by explicitly estimating trial-to-trial fluctuations in confidence criteria (see Suppl. Figure 7 for an explorative approach to estimate confidence criteria fluctuations using hMFC).

One important consequence of criterion fluctuations is that they influence well-established measures of perceptual sensitivity, such as d' ¹. Simulations show that the presence of criterion fluctuations leads to an underestimation of d' (Suppl. Figure 8A-C). Interestingly, a similar pattern emerges for metacognitive sensitivity (*meta-d'*), which is likewise underestimated with strong criterion fluctuations. Because d' and *meta-d'* are affected to a comparable extent, their ratio (*M-ratio*, commonly used as a measure of metacognitive efficiency;³⁴ remains largely unchanged. Consequently, *M-ratio* appears relatively robust to criterion fluctuations. Remarkably, while simulations show that d' is underestimated in the presence of criterion fluctuations, simulated accuracy remains unchanged (Suppl. Figure 8D-E). Therefore, criterion fluctuations are able to explain dissociations between confidence and

accuracy. Such dissociations have been observed in many studies and are often thought to provide a hint about the underlying computations of decision confidence ^{35–40}. Given that confidence accuracy dissociations are so widespread in the literature, it remains beyond the scope of the current project to examine to what extent these different phenomena can be explained by criterion fluctuations, and is therefore left for future work.

The pattern of increased confidence combined with reduced perceptual sensitivity has been associated with a symptom dimension characterized by compulsive behaviors and intrusive thoughts ^{41,42}. When accuracy is used instead of d' , this relationship is no longer observed ⁴². From a theoretical perspective, one explanation for these findings is that individuals scoring higher on these psychiatric symptoms exhibit stronger criterion fluctuations (i.e., increased α and/or σ), leading to apparent reductions in sensitivity and overconfidence. Going beyond single-trial criterion estimates, future work could thus investigate whether individual autoregression parameters are correlated with subclinical variability.

Another important consequence of quantifying confidence as the distance between the decision variable and a fluctuating criterion, is that confidence will naturally become autocorrelated across trials. This phenomenon has been described previously in empirical data, and is often referred to as the confidence leak ⁴³. To explain autocorrelation in confidence (i.e., confidence leak) Rahnev and colleagues (2015) ⁴³ proposed that participants dynamically update their confidence criterion on each trial depending on whether their confidence on the previous trial was higher or lower compared to average. While fluctuations in criterion theoretically might seem an alternative explanation, we did not find strong evidence for this in our empirical data. Although simulations from our model display very strong autocorrelation of confidence (i.e., a strong confidence leak), this effect disappears when adding the estimated criterion fluctuations to the model. However, when applying the same approach to empirical data from Shekhar and Rahnev (2021) ⁶, the effect of previous confidence on current confidence was largely unaffected by inclusion of criterion state. From this we can conclude that at the moment there is no convincing evidence that criterion fluctuations are an alternative explanation to the confidence leak in empirical data.

At the same time, the interpretation of these fluctuations is not uniquely determined. In the current work, we interpret the output of hMFC as reflecting fluctuations in the decision criterion. Alternatively, the same pattern could arise from simultaneous fluctuations in the underlying stimulus distributions while the criterion itself remains static. For example, fluctuations in neuronal excitability could change the baseline firing rate of neurons over time, effectively shifting the internal evidence distributions ⁴⁴. From a modeling perspective, these

alternatives are mathematically equivalent, and distinguishing between them would require additional neural data.

A further limitation of hMFC is that it is formulated within the Signal Detection Theory (SDT) framework, which does not account for reaction times. Explaining reaction time distributions typically requires sequential sampling models such as the Drift Diffusion Model (DDM) ⁴⁵. Importantly, whereas SDT includes a single bias parameter (the criterion), the DDM includes two potential sources of bias: the starting point and the drift bias ¹⁴. To investigate the mapping between these parameters and the SDT criterion, we developed a DDM variant where starting point and drift bias fluctuate according to an AR(1) process (see also Vloeberghs et al., 2026) ⁴⁶. We fitted this DDM to the same empirical datasets as hMFC, and found that criterion fluctuations were significantly predicted by drift bias, both when analyzing all datasets together or individually (Suppl. Figure 9). Criterion fluctuations also mapped onto fluctuations in starting point when analyzing all datasets together. When analyzing the datasets individually, the effect was only present in 4 out of 14 datasets. Note that one dataset was excluded from the analysis due to a task design that was incompatible with the DDM. These results suggest that drift bias can be consistently detected at the dataset level, whereas starting point is considerably noisier and requires substantially more data to be reliably identified. Taken together, we can conclude that criterion fluctuations measured by the hMFC toolbox reflect both fluctuations in drift bias and starting point.

An important question for future work is how these findings relate to alternative conceptualizations of the decision criterion. The current paper considers the criterion in evidence space, whereas Bayesian models define the criterion in probability space. Future research could compare these approaches and examine whether they allow for a clearer distinction between the distance-to-criterion account and Bayesian accounts of confidence ⁸.

In summary, our study provides robust evidence that confidence is modulated by trial-to-trial fluctuations in the decision criterion. This finding shows that variability in confidence judgments, which is often assumed to reflect noise, reflects genuine computation of confidence as distance-to-criterion. With this work new insights are provided into the computational underpinnings of decision confidence.

Methods

Open science statement

A first exploratory analysis was performed on the data of Experiment 4 in Shekhar and Rahnev (2021) which is publicly available. Afterwards, we performed a pre-registered confirmatory analysis of 14 datasets from the Confidence Database (Rahnev et al., 2021). The pre-registration can be found at <https://osf.io/mxuv2>. For ease of reading, in the remainder we will jointly discuss the results from the exploratory analysis together with the 14 confirmatory analyses. Finally, a pre-registered analysis regarding individual differences can be found in Supplementary Analysis 3. The analyses on the two additional datasets involving reaction times, pupil size, and the neural marker of confidence is not included in the pre-registration.

Simulation study

To investigate how criterion fluctuations shape confidence, we simulated 6000 trials from an SDT observer. On each trial, a decision variable was sampled from a normal distribution with mean $\mu = [3, -2, -1, 1, 2, 3]$ and standard deviation $\sigma = 1$. The decision criterion varied across trials according to a first-order autoregressive process, AR(1), with autoregressive coefficient $\alpha = .9995$ and noise standard deviation $\sigma = .05$. A binary response on each trial was obtained by comparing the decision variable to the decision criterion. Confidence was then quantified as the absolute distance between the decision variable and the decision criterion. To examine the relationship between confidence and criterion fluctuations, we used a linear regression model where confidence was predicted by a full-factorial model including as predictors the decision criterion, stimulus direction, and stimulus strength. Stimulus strength was defined as the absolute value of the mean μ of the normal distribution, while stimulus direction corresponded to its sign. For visualization purposes only, the decision criterion was divided into equally sized bins.

Dataset selection

As pre-registered, we only included studies that met the following inclusion criteria: 1) participants reported trial-by-trial confidence using a confidence rating scale, 2) each participant completed a minimum of 500 trials (i.e., to allow robust estimation of trial-by-trial criterion with hMFC), 3) the main manipulation involved the stimulus strength presented to participants on the screen, and 4) no trial-by-trial feedback was provided. Based on these criteria, a total of 14 datasets were selected (i.e., in addition to Shekhar & Rahnev (2021))⁶

which was used for the exploratory analysis). Note that in contrast to the pre-registration the datasets of Yeon et al. (experiment 1, unpublished) and Clark et al. (2018) ⁴⁷ are not included in the current analysis. The former was wrongly classified and only used one level of stimulus strength, and for the latter the fitting procedure of hMFC did not converge due the low number of subjects (n=4).

From the fifteen studies included, eight used Gabor patches as experimental stimuli, four studies used the random dot motion task (i.e., participants indicated whether they perceived left- or rightward movement within a cloud of moving dots), two studies asked participants to decide whether the left or right panel contained the most dots, and one study used ambiguous faces, which had to be classified as either looking fearful or happy. The majority of the studies, 10 out of 15, used a discrete 4-point confidence scale, whereas the other five studies either used a continuous scale or a 2-, 3-, or 6-point scale. Across studies, the number of subjects varied from 10 to 54, and the number of trials ranged between 504 and 3240. Thirteen subjects were excluded because they either performed at chance level or provided the same confidence rating on over 90% of trials. Six additional subjects were removed due to estimated criterion states larger than 10. Values this large saturate the sigmoid-transformed log-odds in hMFC, thereby jeopardizing the accuracy of the model estimates. In total, after excluding these subjects the final datasets comprised 379 subjects with 459,091 trials in total. Detailed information about each of the studies can be found in Supplementary Table 4.

To examine whether our findings extend to implicit measures of confidence and to neurophysiological correlates of confidence, we analyzed two additional datasets. For reaction time and pupil size, we used data from Urai et al. (2017) ⁷, which included 27 human observers performing 500 trials of a two-interval forced choice (2-IFC) motion coherence discrimination task. In this task, participants judged whether a test random dot motion stimulus had higher or lower coherence than a reference stimulus, while their pupil sizes were recorded. Throughout the experiment, five stimulus strength levels were presented. Note that this dataset did not include confidence ratings. Reaction times were log-transformed and standardized. Pupil size was calculated as the mean baseline-corrected signal during the 250ms preceding feedback. More details on pupillometry preprocessing are provided in Urai et al. (2017) ⁷. To assess the modulation of criterion fluctuations in a neural marker of confidence (error positivity), we reanalyzed data from Boldt & Yeung (2015) ²⁷. In this study, 16 participants completed 864 trials in which they judged which of two panels contained more dots and then reported their confidence on a six-point scale. Only a single stimulus strength level was used in this experiment. Single-trial error positivity amplitudes were calculated at Pz in a time window of

250ms to 350ms post-response, following Boldt & Yeung (2015) ²⁷. Technical details of the EEG preprocessing are available in the original study.

Estimating the single-trial decision criterion using hMFC

The single-trial decision criterion was obtained by fitting the Hierarchical Model for Fluctuations in Criterion (hMFC) separately to each of the 15 datasets ²⁰. This model uses a Bernoulli distribution to predict the binary response y_t at trial t :

$$y_t \sim \text{Bern}(f(\mathbf{w}^T \mathbf{u}_t + x_t)) \quad (1)$$

where the probability of a response is determined by a set of input variables $\mathbf{u}_t$, weighted by $\mathbf{w}$, and a latent variable x_t representing the decision criterion for that trial. The sigmoid transformation $f(z)$ restricts the probability between 0 and 1. The criterion state x_t is modelled as a first-order autoregressive model with intercept b , autoregressive coefficient a , and error variance σ^2 :

$$x_t \sim b + a x_{t-1} + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2) \quad (2)$$

Adding the fluctuating value x_t to the Bernoulli log-odds is equivalent to comparing a decision variable to a fluctuating criterion. Note, however, that the estimated criterion x_t , as outputted by hMFC, is a bias on the probability of making a certain response (e.g., *right* response). Thus, a positive criterion state will increase the probability of a *right* response, and therefore corresponds in SDT to a criterion shifted to the *left*. For ease of interpretation, in further analyses the sign of x_t is inverted such that its interpretation of x_t is in line with the traditional SDT interpretation. Note that systematic effects on the criterion (e.g., stimulus base-rate manipulations, feedback conditions, or response on previous trial) can be estimated by including this variable as covariate $\mathbf{u}_t$ in hMFC. The AR(1) process for the criterion fluctuations therefore models the endogenous fluctuations on top of the systematic effects on the criterion of the included covariates. Although these criterion fluctuations are modeled as an AR(1) process, hMFC can recover deviations from this assumption through its use of the Forward Filtering/Backward Sampling algorithm ⁴⁸. For instance, capturing the influence of omitted variables or estimating non-stationary patterns in the criterion dynamics, such as systematic trends over time (Suppl. Figure 10).

In addition to the per-subject parameters $(\mathbf{w}, b, a, \sigma^2)$, hMFC estimates hierarchical (group-level) distributions, allowing to share statistical strength across subjects to enhance the estimation accuracy of these individual parameters. Through this hierarchical structure, hMFC can accurately recover the model parameters, even with as few as 500 trials per subject. Most

importantly, hMFC shows excellent recovery of the latent decision criterion trajectories (Figure 2A,B,C), providing accurate trial-to-trial estimates of the decision criterion. For full details on the model and its parameter recovery, we refer the reader to Vloeberghs et al. (2025) ²⁰.

In the current study, as input variables $\mathbf{u}_t$ for hMFC we considered current trial stimulus, previous trial stimulus, and both response and confidence on the previous trial, together with their interaction. Note that in the data of Urai et al. (2017) ⁷, used to investigate pupil size, self-reported confidence rating were absent and thus confidence on the previous trial could not be used as a predictor in hMFC. The model was run with four chains, each generating 5000 posterior samples with the first 500 samples discarded as burn-in. Convergence of the chains was assessed using the R-hat diagnostic. Fitting the full model with all the hierarchical (group-level) parameters resulted in convergence issues of the different chains (i.e., R-hat > 1.05) for the majority of the datasets. These issues were resolved by fixing μ_{σ^2} and β_{σ^2} , the two parameters of the inverse gamma hierarchical prior for σ^2 , to the estimated values for Shekhar & Rahnev (2021) ⁶, for which the model did properly converge ($\mu_{\sigma^2} = 0.15$ and $\beta_{\sigma^2} = 0.15$). For consistency, all datasets were fitted with these two hyperparameters fixed. To ensure the robustness of our main results in light of these fixed parameters, we conducted a sensitivity analysis across a range of values for μ_{σ^2} and β_{σ^2} (Suppl. Table 3).

Predicting confidence with single-trial criterion state

To test the hypothesis that criterion fluctuations shape confidence ratings, we estimated three linear mixed effects models. The dependent variable in each of these models was reported confidence, which was rescaled to range between 0 to 1 for all datasets. In the first model, confidence was predicted based on the interaction between stimulus strength, stimulus direction, and the single-trial criterion estimates, obtained with hMFC. In a random dot motion task, stimulus strength would correspond to the motion coherence and stimulus direction would refer to left- or rightward movement. Stimulus strength was recoded in each dataset to fall between 0 and 1, and stimulus direction was recoded as -1 and 1. The second model only includes stimulus strength and stimulus direction. The third model includes the estimated criterion trajectories but shuffled *across* subjects within each dataset (cf. session permutation method ²⁴). Statistically comparing autocorrelated timeseries can introduce spurious significant effects when not sufficiently controlled for. To ensure that our models only picked up on confidence variability that are truly driven by trial-to-trial criterion fluctuations, we therefore employed the session permutation method by comparing Model 1 (original criterion fluctuations) with Model 3 (shuffled criterion fluctuations). If confidence ratings are indeed

influenced by criterion fluctuations, the first model including trial-by-trial criterion estimates should be favored over the other two models in a model selection procedure.

Model 1: Confidence $\sim$ stimulus strength * stimulus direction * criterion + (1 + stimulus strength + criterion | study / subject)

Model 2: Confidence $\sim$ stimulus strength * stimulus direction + (1 + stimulus strength | study / subject)

Model 3: Confidence $\sim$ stimulus strength * stimulus direction * shuffled criterion + (1 + stimulus strength + stimulus direction + shuffled criterion | study / subject)

First, we estimated the three models to all datasets *combined*. To account for the nesting of participants within studies when analyzing the datasets simultaneously, we used nested random effects. For model selection, it should be noted that the comparisons between Model 1 and Model 2, and between Model 1 and Model 3, were not conducted on the same dataset. Therefore, the results of these comparisons should not be compared directly. Specifically, the session permutation method used for Model 3 required all subjects within a dataset to have the same number of trials. To achieve this, within each dataset, we determined the cutoff based on the subject with the fewest trials and excluded any trials exceeding this count. As such, the estimated criterion trajectories could be shuffled *across* subjects. While the comparison between Model 1 and Model 3 was conducted on this truncated dataset, the comparison between Model 1 and Model 2 was performed on the full dataset. For completeness, Supplementary Analysis 1 reports the comparison between Model 1 and Model 2 using the truncated dataset as well. Second, in addition to analyzing all datasets combined, we analyzed each dataset *separately* using Model 1, but without the random effects nested within each study.

To examine whether fluctuations in decision criterion influence reaction times and neurophysiological markers of confidence, we applied the same approach described above, replacing confidence ratings with reaction times, pupil size, or error positivity (Pe) amplitude as the dependent variable.

For all models, the random effects structure was obtained using an automated procedure which fits all possible random effect structures in parallel and selects the model with the lowest AIC. All statistical analyses were performed in R (4.6.0) using Rstudio (2026.05.0)⁴⁹. The linear mixed effects models were fitted using the lme4 package (1.1.37)⁵⁰. The ordinal mixed regression models were fitted with the ordinal package (2022.11.16)⁵¹.

Data Availability

All data analyzed in the current study can be found at www.github.com/robinvloeberghs/Confidence-and-criterion-fluctuations.

Code Availability

The code for all analyses is available at www.github.com/robinvloeberghs/Confidence-and-criterion-fluctuations.

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Author Contributions

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Competing Interests

The authors declare no competing interests.

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Supplementary Analysis 1

To ensure full comparability, we additionally fitted the model without criterion fluctuations (Model 2) to the truncated dataset. We then compared it with the two alternative models fitted to the same truncated dataset: the model including criterion fluctuations (Model 1) and the model with shuffled criterion fluctuations (Model 3; cf. session permutation method).

Consistent with the model selection on the full dataset, we found that the model including criterion fluctuations (Model 1) outperforms the model without criterion fluctuations (Model 2) ($\Delta AIC = -11898$, $\Delta BIC = -11457$, $\chi^2(40) = 11978$, $p < .001$). When comparing Model 2 with the model including the shuffled criterion fluctuations (Model 3), we found that the simpler Model 2 is clearly preferred ($\Delta AIC = -48890$, $\Delta BIC = -48934$).

Supplementary Analysis 2

We performed the analysis of reaction times, pupil and Pe amplitude for correct and error trials separately. First, for reaction times, when re-analyzing reaction times for correct trials we find a significant interaction between stimulus direction and criterion ($\beta = 0.410$, $SE = 0.022$, $F(1,50058) = 333.919$, $p < .001$). Specifically, for a right stimulus, longer reaction times are observed when the criterion fluctuates towards the right. This effect is also observed when analyzing the full dataset. All VIFs were below 5.431. For error trials, we find that the pattern flips ($\beta = -0.335$, $SE = 0.038$, $F(1,17085) = 76.432$, $p < .001$), with now shorter reaction times for right stimulus with a criterion shifted to the right. For these error trials with a right stimulus, the response was left, therefore the decision variable fell on the left side of the criterion. Hence, a criterion fluctuating to the right increases the distance to the decision variable and should result in higher confidence, explaining the shorter reaction times. All VIFs were below 6.236.

Next, for pupil size we find the same pattern with a significant interaction between stimulus direction and criterion for correct trials ($\beta = 0.166$, $SE = 0.029$, $F(1,2560) = 32.590$, $p < .001$) and error trials, where the sign flips ($\beta = -0.132$, $SE = 0.044$, $F(1,17089) = 8.838$, $p = .003$). All VIFs were below 6.212.

Lastly, predicting the Pe amplitude for correct trials we find that the interaction between stimulus direction and criterion is in the expected direction, but fails to reach significance ($\beta = 0.237$, $SE = 0.230$, $F(1,138.43) = 1.069$, $p = .303$). For the error trials the interaction is significant, and as expected switched sign ($\beta = -1.031$, $SE = 0.477$, $F(1,2330.9) = 4.674$, $p = .031$). For both models, no multicollinearity issues were detected (all VIFS < 3.797).

Supplementary Analysis 3

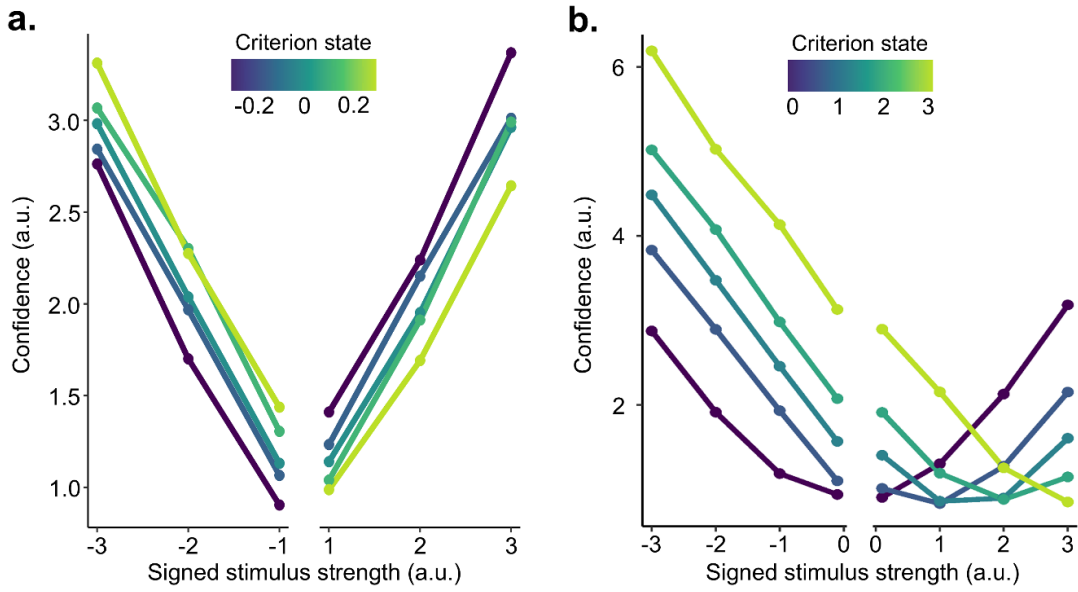
In a pre-registered exploratory analysis we investigated the relationship between confidence variability, measured as the standard deviation (SD) for each subject, the variability of the criterion fluctuations, and the hMFC parameters, α and σ^2 , which capture the temporal dynamics of the criterion fluctuations. All predictor variables were standardized.

Model A: Confidence SD $\sim \alpha * \sigma^2 + (1 | \text{study})$

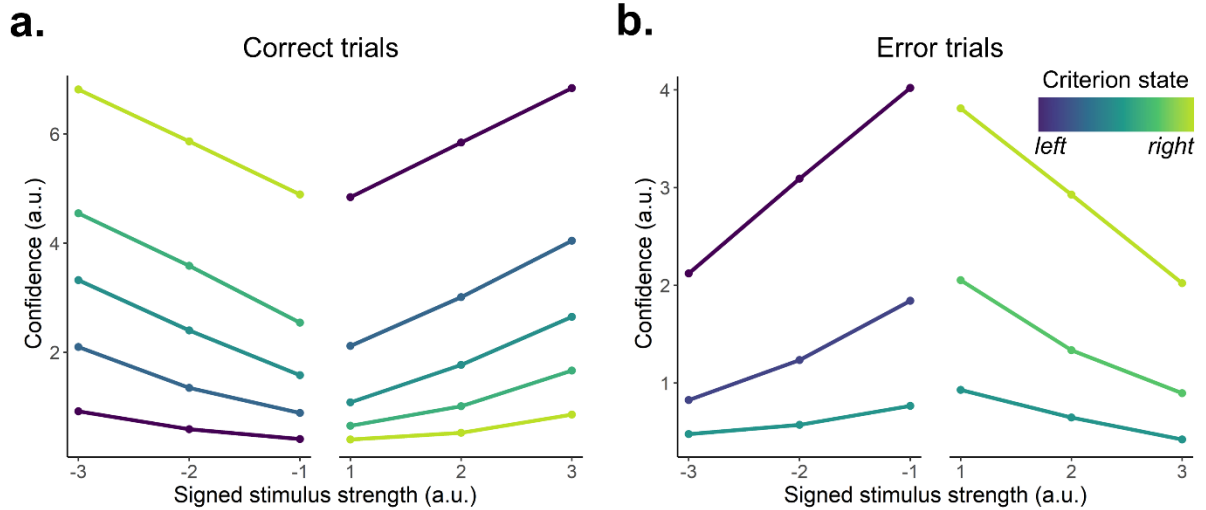
Model B: Confidence SD $\sim \text{criterion fluctuations SD} + (1 | \text{study})$

In Model A the main effect of α was significant ($\beta = -0.012$, $SE = 0.005$, $F(1,373) = 7.020$, $p = .008$), suggesting that a higher value for α corresponds to lower variability in confidence. Both the effects of σ^2 and the interaction between α and σ^2 were not significant (all $p \geq .323$). In Model B we found a significant main effect for the variability of the criterion fluctuations ($\beta = -0.008$, $SE = 0.003$, $F(1,368) = 5.788$, $p = .017$). This indicates that subjects with larger variability in criterion showed less variability in their confidence ratings. No multicollinearity problems were detected (all VIF < 1.013). One possible explanation for these results is that subjects with high variability in criterion may be less engaged and attentive, and more often give the same confidence rating throughout the task. Note that including the per-subject criterion trajectory mean μ_x , which is used in hMFC to estimate the intercept b , in Model A does not change the results. Therefore, we report the pre-registered analysis without μ_x .

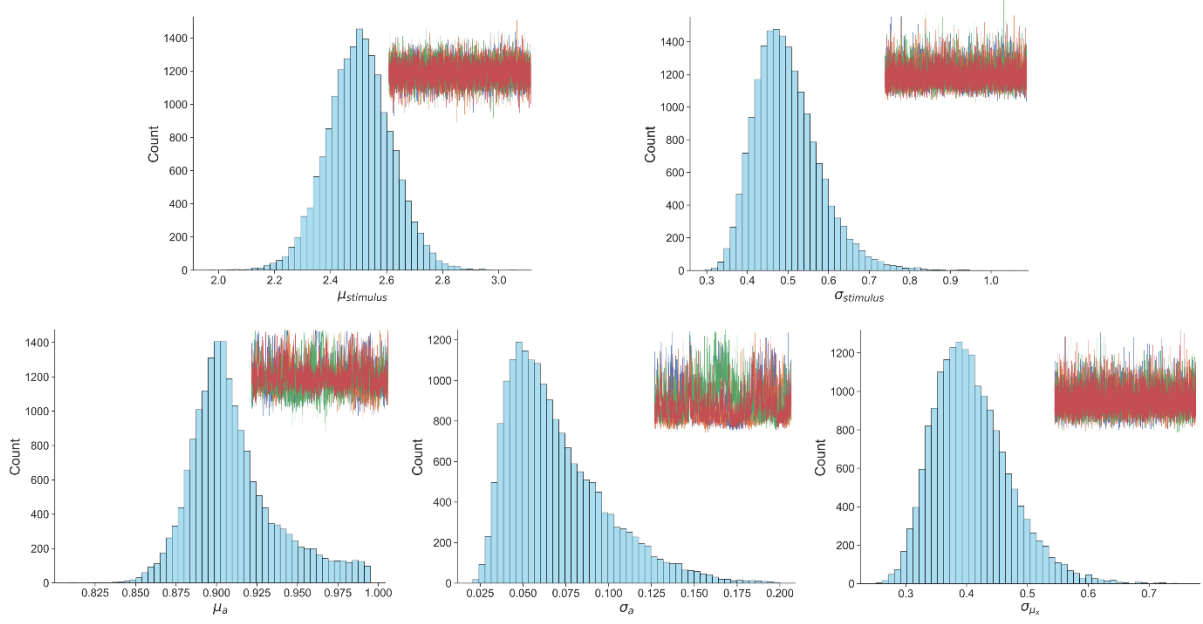
Supplementary figures



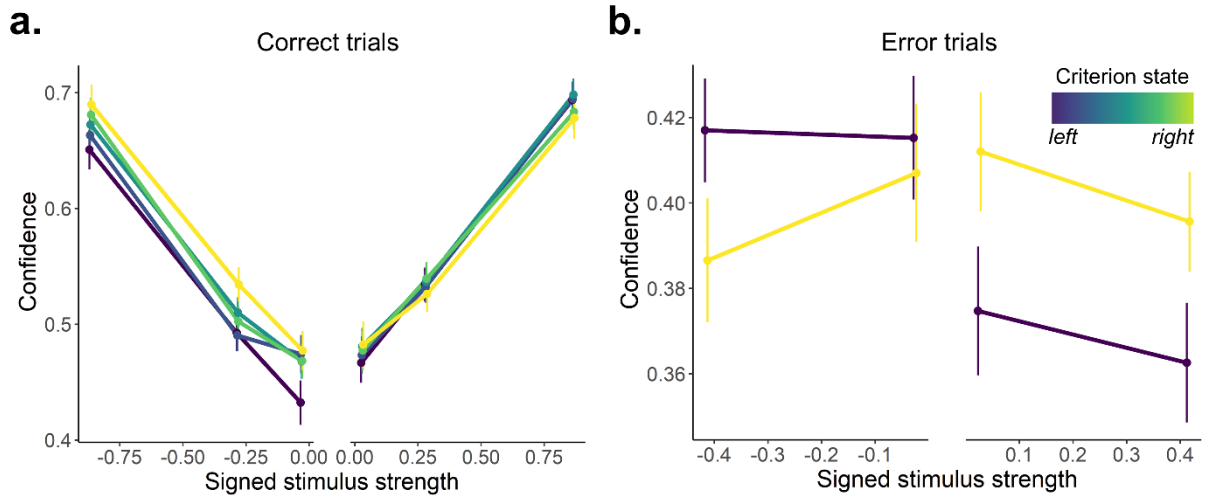
Supplementary Figure 1: Investigating the three-way interaction between stimulus direction, stimulus strength, and criterion. We simulated an agent with 6000 trials from the signal detection theory (SDT) framework, with confidence defined as the absolute distance between the decision variable and the decision criterion. The criterion fluctuated across trials according to an AR(1) process. The values on the x-axes represent the mean of the generative SDT stimulus distribution **A)** When predicting confidence using linear regression, the interaction between stimulus direction and criterion was significant ($\beta = -1.297$, $SE = 0.264$, $F(1) = 283.350$, $p < .001$), with higher confidence for leftward stimuli when the criterion fluctuates towards the right. However, the three-way interaction between stimulus direction, stimulus strength, and criterion was not significant ($\beta = -.188$, $SE = 0.122$, $F(1) = 2.358$, $p = 0.125$). This occurs because the lines representing different criterion states are parallel rather than curved, as in Figure 1C. Whether this three-way interaction emerges depends on the combination of the SDT stimulus distributions and the criterion. Confidence is lowest on trials where the criterion lies near the center of the stimulus distribution. If such trials occur in the dataset, the resulting pattern produces curved lines and consequently a significant three-way interaction. In this simulation, the criterion fluctuated within a narrow range (-0.2 to 0.2), which did not overlap with the means of the SDT stimulus distributions. As a result, the lines remain parallel and the three-way interaction is not significant. Taken together, these simulations show that the three-way interaction between stimulus direction, stimulus strength, and criterion is not a reliable signature of confidence reflecting distance to a fluctuating criterion. Instead, it arises from the specific combination of stimulus configuration and the range of criterion fluctuations. **B)** To illustrate that confidence on average is lowest when the criterion coincides with the stimulus mean, we simulated an agent whose criterion fluctuates exclusively on the right side (i.e., only positive criterion values). Considering rightward stimuli (positive x-axis values), the lowest confidence for each stimulus strength—corresponding to the mean of the rightward SDT stimulus distribution—occurs when the criterion equals that stimulus strength. In other words, confidence is lowest when the criterion lies at the center of the stimulus distribution.



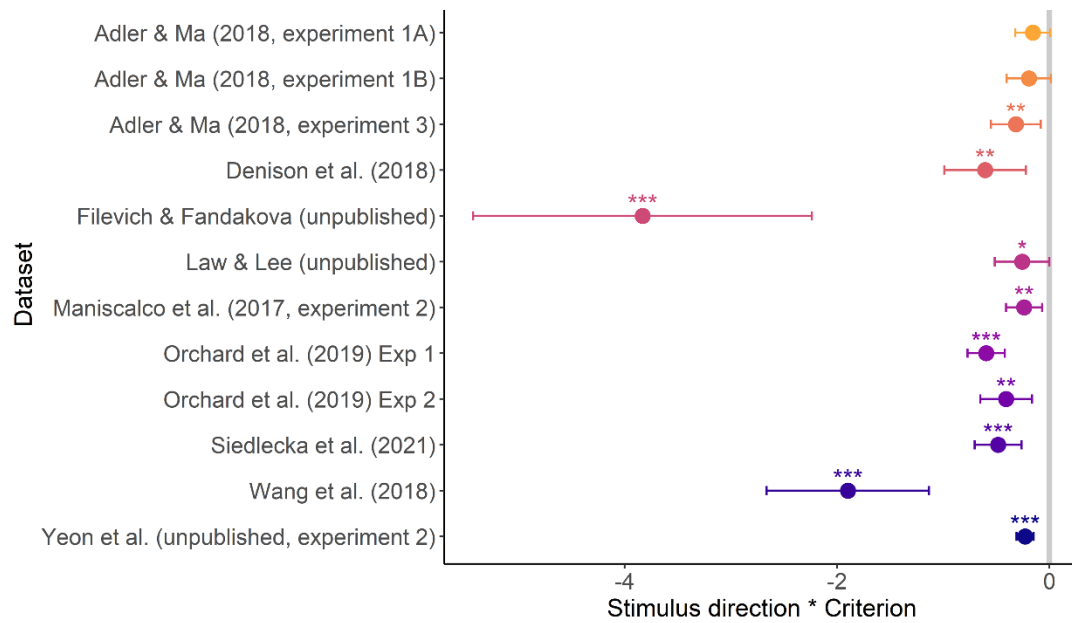
Supplementary Figure 2: Confidence as a function of stimulus direction and stimulus strength, separately for correct and errors trials in simulated data. Data were generated from a signal detection theory (SDT) framework in which the decision criterion fluctuated across trials according to an AR(1) process (6000 trials). **A)** When analyzing correct trials (left panel), the same pattern emerges as in the full dataset (Figure 1C). For leftward stimuli (negative values on x-axis), confidence is higher when the criterion fluctuates to the right compared to the left. For rightward stimuli (positive values on x-axis), this pattern reverses, with higher confidence when the criterion fluctuates to the left compared to the right. In the full regression model including stimulus direction, stimulus strength, and criterion, we find a significant interaction between stimulus direction and criterion ($\beta = -1.747$, $SE = 0.012$, $F(1) = 196945.22$, $p < .001$), as well as a significant three-way interaction between stimulus direction, stimulus strength, and criterion ($\beta = -0.04$, $SE = 0.005$, $F(1) = 59.40$, $p < .001$). **B)** While for the correct trials confidence increases as the stimulus strength increases, for the errors trials confidence decreases (right panel), consistent with the statistical signatures of confidence (Sanders et al., 2016). Additionally, the interaction between stimulus direction and criterion reverses relative to correct trials ($\beta = 1.668$, $SE = 0.02$, $F(1) = 29617.62$, $p < .001$). Specifically, for leftward stimuli, confidence is highest when the criterion fluctuates to the left. This occurs because, on error trials for leftward stimuli, the response was “right,” implying that the decision variable was situated on the right side of the criterion. A leftward criterion shift increases the distance between the decision variable and the criterion, thereby increasing confidence. The opposite pattern is observed for rightward stimuli. Finally, as with correct trials, the three-way interaction is also significant for error trials ($\beta = -0.114$, $SE = 0.011$, $F(1) = 102.21$, $p < .001$).



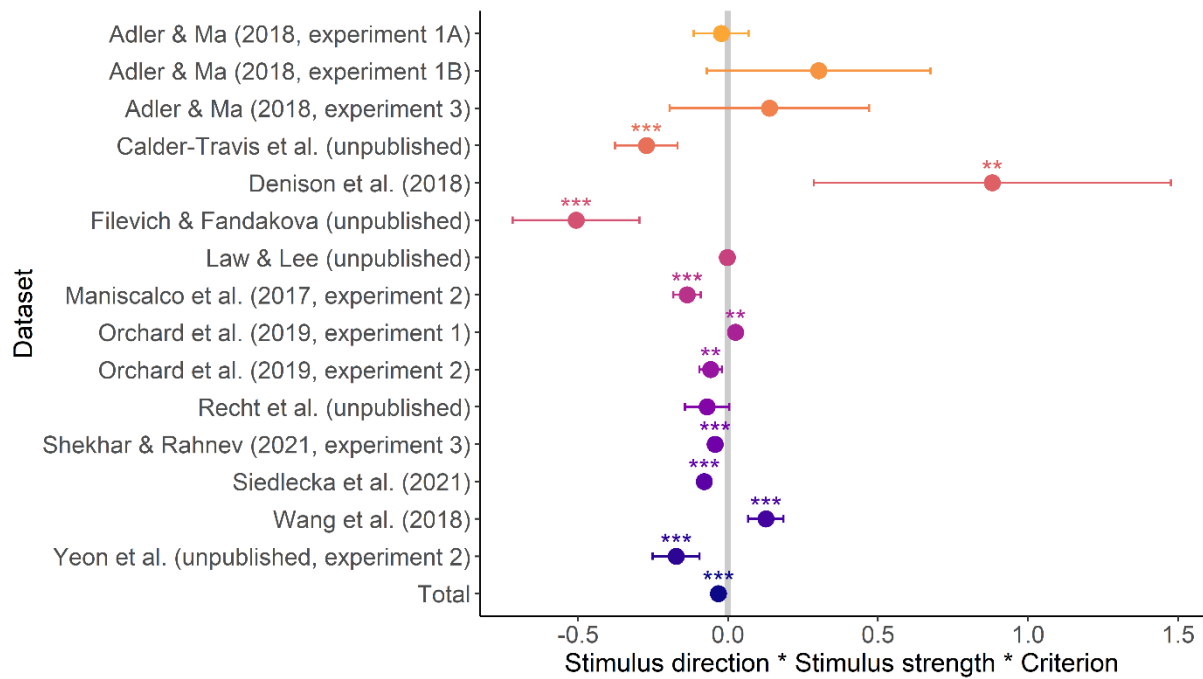
Supplementary Figure 3: The posterior distributions of the hMFC hierarchical (group-level) parameters together with trace plots for Shekhar & Rahnev (2021) ⁶. For each input variable w hMFC assumes a hierarchical (group-level) normal distribution with two estimated parameters: μ_w and σ_w . For the autoregressive coefficient a a truncated normal distribution constrained to $[0, 1]$ is specified with parameters μ_a and σ_a . Lastly, the per-subject criterion trajectory mean μ_x , which is estimated and allows a closed-form estimation for the intercept b , follows a zero-mean normal distribution with σ_{μ_x} as standard deviation (see Vloeberghs et al. (2025) ²² for more technical details). All parameters for each dataset have a R -hat lower than 1.05, which suggests a proper mixing of the chains of the Gibbs sampler. Note that for μ_w and σ_w the highest R -hat is shown out of the five input variables.



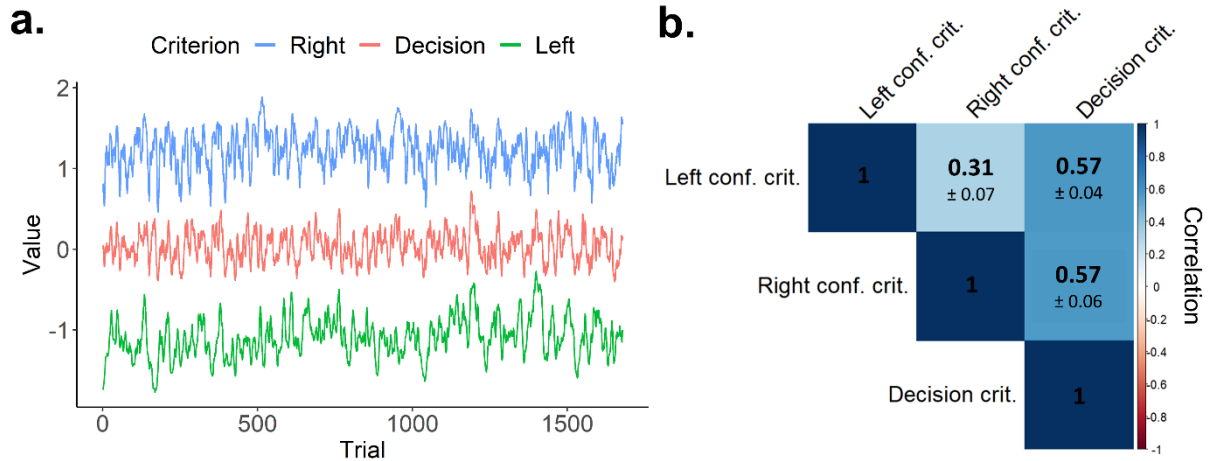
Supplementary Figure 4: Confidence as a function of stimulus direction and stimulus strength for correct and errors trials in empirical data. Data are shown across the 15 datasets. Error bars reflect standard errors. For plotting purposes only, criterion states were binned and the average within each bin is shown. For the error trials we reduced the number of bins due to lower trial counts. **A)** The analysis of correct trials using linear mixed models showed a significant two-way interaction between stimulus direction and criterion ($\beta = -0.035$, $SE = 0.003$, $F(1,17988.10) = 148.862$, $p < .001$), with higher confidence for left stimuli when the criterion fluctuates towards the right. Also the three-way interaction was significant ($\beta = 0.011$, $SE = 0.005$, $F(1,4002.60) = 5.133$, $p = .024$), but in the opposite direction than expected from simulations. In a model selection we found that the model including criterion fluctuations (Model 1) outperformed the model without criterion fluctuations (Model 2) ($\Delta AIC = -8745$, $\Delta BIC = -8445$, $\chi^2(28) = 8800$, $p < .001$). In addition, the model with the original criterion fluctuations (Model 1) was preferred over the model including the shuffled criterion fluctuations (Model 3, cf. session permutation method) ($\Delta AIC = -712$, $\Delta BIC = -712$). **B)** When analyzing the errors trials we find a significant interaction between stimulus direction and criterion ($\beta = 0.048$, $SE = 0.011$, $F(1,10) = 17.689$, $p = .002$). In line with the simulations, we find that the pattern is reversed compared to the correct trials, with higher confidence for left stimulus when the criterion fluctuates towards the left. The three-way interaction was not significant ($\beta = -0.004$, $SE = 0.009$, $F(1,9564) = 0.170$, $p = .680$). Similar to the analysis of the correct trials, the model including criterion fluctuations (Model 1) outperformed the model without (Model 2) ($\Delta AIC = -1016$, $\Delta BIC = -802$, $\chi^2(22) = 1060$, $p < .001$). Also the model with original criterion fluctuations (Model 1) was preferred over the model with the shuffled criterion fluctuations (Model 3) ($\Delta AIC = -1011$, $\Delta BIC = -797$).



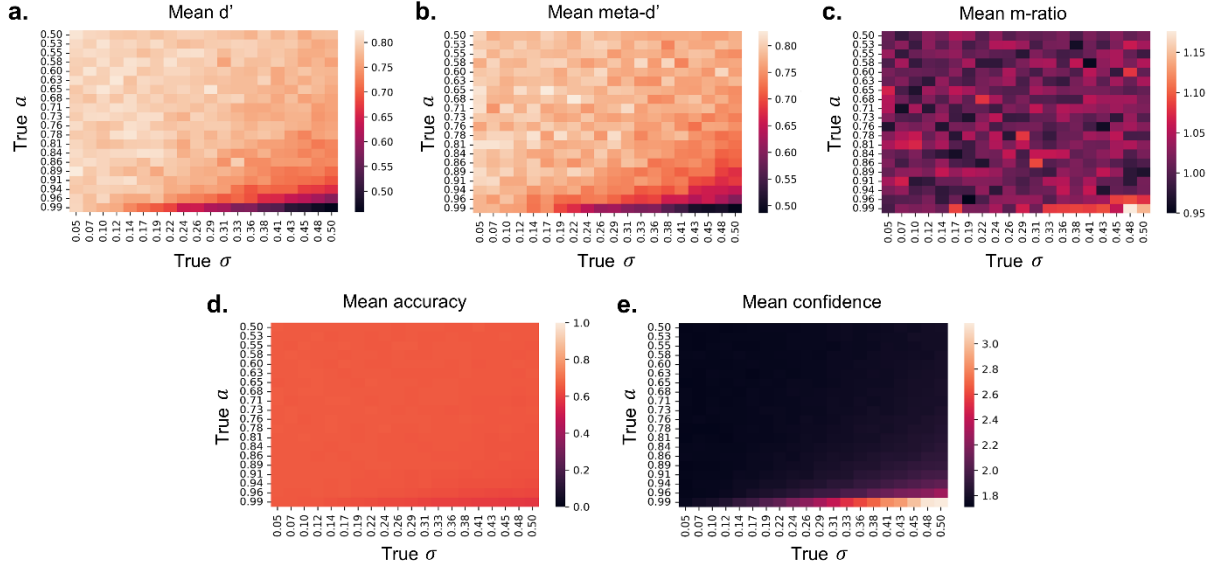
Supplementary Figure 5: Interaction between stimulus direction and criterion when analyzing datasets with an ordinal mixed model. *To investigate the robustness of our results beyond linear mixed models, we re-analyzed the data with a cumulative-link mixed model from the ordinal package in R . We selected the datasets with confidence on a 3-, 4- or 6-point rating scale, resulting in 12 datasets in total. Predicting confidence, the crucial interaction between stimulus direction and criterion was significant for 10 out of 12 datasets. Two datasets, which showed a significant interaction using linear mixed models, are now not significant (Adler & Ma, 2018; experiment 1A: $p = 0.061$, experiment 1B: $p = 0.066$). We therefore conclude that our results are robust for different types of regression models.*



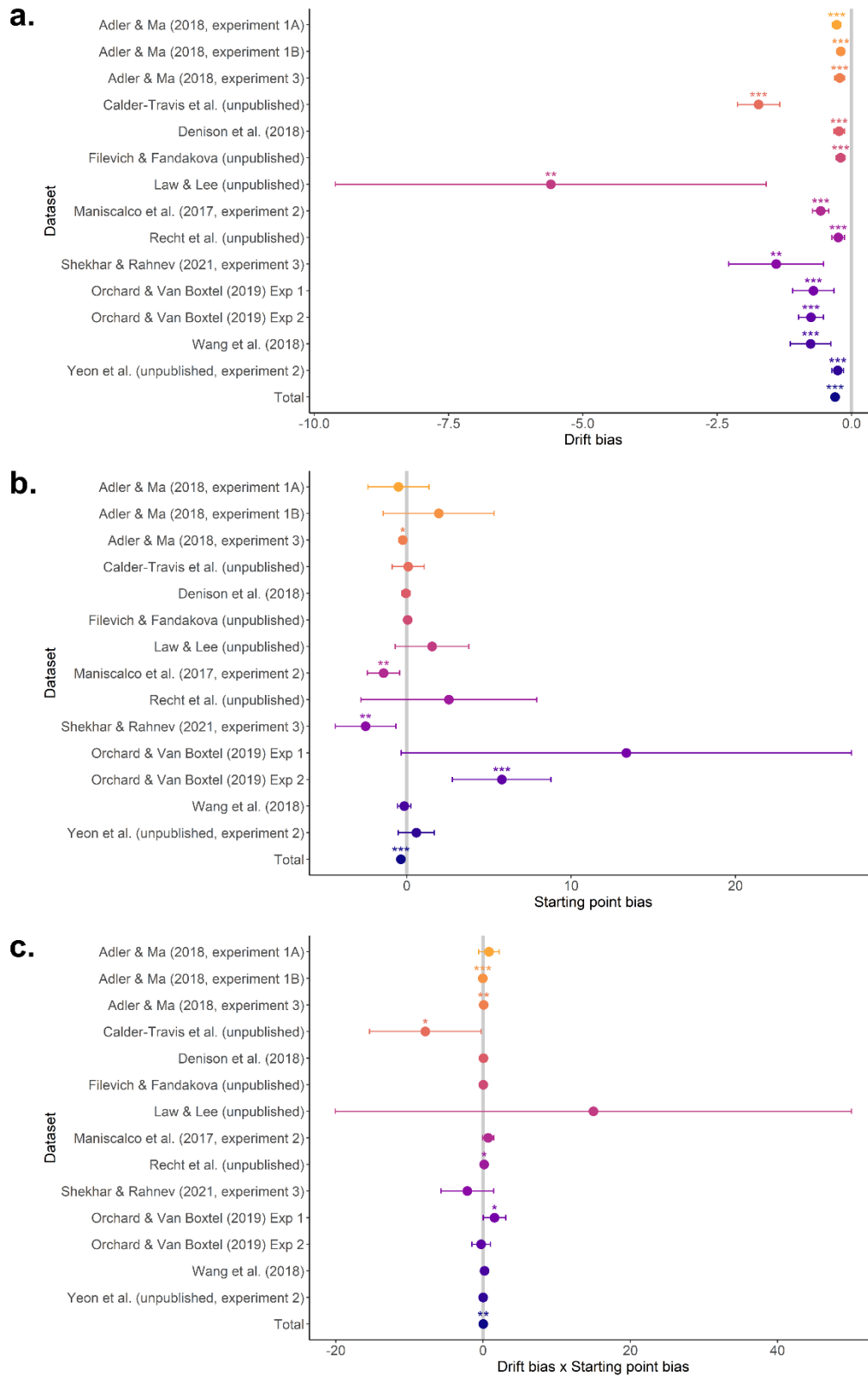
Supplementary Figure 6: The three-way interaction of stimulus direction, stimulus strength, and criterion for each dataset separately. The parameter estimates and their 95% confidence intervals are shown obtained from linear mixed models fitted separately to every dataset. The estimate plotted at the bottom, labelled “Total”, shows the parameter estimate from the model fitted to all datasets (taking into account the nesting of subjects in experiments). Note: * = $p < .05$, ** = $p < .01$, *** = $p < .001$.



Supplementary Figure 7: Using hMFC to estimate both decision criterion and confidence criteria in empirical data of Recht et al. (unpublished). In a first explorative attempt we used hMFC to measure confidence criteria. Specifically, in a dataset with two confidence options (low vs. high), and thus two confidence criteria (for left and right responses), we recoded right response with high confidence as one, and the rest as zero. Under this transformation, the output of hMFC should represent fluctuations in the right confidence criterion. For the left confidence criterion, all data is recoded to one, except for left high confidence responses which were coded as zero. Thus, to estimate two confidence criteria hMFC is fitted two times. While this approach is highly explorative and requires additional testing, we applied this approach as a proof of concept to an empirical dataset which features a binary confidence scale, i.e., data of Recht et al. (unpublished), consisting of 10 subjects with 1679 trials each. **A)** The estimated posterior means of the decision criterion, and left and right confidence criteria for one example subject. Across subjects, the confidence criteria show clear autocorrelated fluctuations (for left confidence criterion: $\mu_\alpha = 0.82$, $\sigma_\alpha = 0.055$, for right confidence criterion: $\mu_\alpha = 0.80$, $\sigma_\alpha = 0.11$) **B)** Both confidence criteria correlate with fluctuations in the decision criterion (for left confidence criterion: $r = 0.57$, $SE = 0.04$, for right confidence criterion: $r = 0.57$, $SE = 0.06$). The confidence criteria themselves also correlate ($r = 0.31$, $SE = 0.07$). Most importantly, when predicting the binary confidence ratings in a logistic mixed model based on the confidence criteria and the interaction between stimulus direction and decision criterion, we find a significant effect of both left confidence criterion ($\beta = 4.241$, $SE = 0.466$, $\chi^2(1) = 82.706$, $p < .001$), and right confidence criterion ($\beta = -6.405$, $SE = 1.123$, $\chi^2(1) = 32.517$, $p < .001$). As the left confidence criterion fluctuates to the right, we see higher confidence, whereas for the right confidence criterion this results in lower confidence. Importantly, in this model we also found the significant interaction between stimulus direction and decision criterion ($\beta = -1.833$, $SE = 0.357$, $\chi^2(1) = 26.315$, $p < .001$), suggesting that fluctuations in decision criterion and in confidence criterion explain independent sources of variability in confidence. No multicollinearity issues were identified (all VIFs < 2.903).

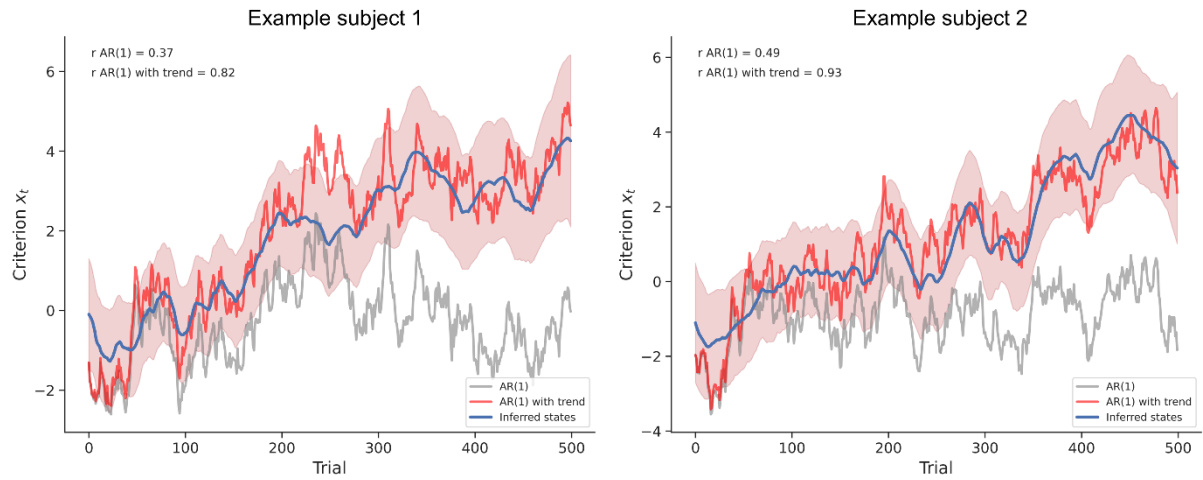


Supplementary Figure 8: The influence of criterion fluctuations on d' , meta- d' , M-ratio, accuracy, and confidence. Data was simulated from the signal detection theory (SDT) framework. Confidence was defined as distance-to-criterion, with the criterion fluctuating from trial-to-trial according to an AR(1) process. We investigated the influence of criterion fluctuations on perceptual sensitivity (d'), metacognitive sensitivity (meta- d'), and metacognitive efficiency (M-ratio), by systematically varying the AR(1) parameters: the autoregressive coefficient α , and the error standard deviation σ . For each parameter combination, we simulated 100 subjects with 500 trials each. **A)** Consistent with the findings of Vloeberghs et al., (2025), we observe that stronger criterion fluctuations (i.e., larger α and σ) lead to greater underestimation of d' (true $d' = 0.80$) when criterion fluctuations are not explicitly accounted for. **B)** A similar pattern of underestimation is observed for meta- d' . **C)** Because meta- d' and d' exhibit similar patterns of underestimation, the effect of criterion fluctuations largely cancels out in M-ratio (meta- d' / d'). Consequently, this measure for metacognitive efficiency remains relatively unaffected by criterion fluctuations. **D)** Average accuracy is not affected by criterion fluctuations. This is because the effect of criterion fluctuations on hits and false alarms cancel out over time. When the criterion fluctuates towards the left, this will lead to a higher hit rate. However, when the criterion then fluctuates towards the right, the false alarm rate will increase. Taken together, these effects on accuracy are cancelled out. Note that it may seem contradicting that accuracy remains unaffected while d' is affected. The reason for this lies in how d' is calculated. Consider the most simple case of criterion fluctuations: half of the trial have a left shifted criterion, the other half a right shifted criterion. The formula for d' effectively takes the averages of the hits and false alarms of the two conditions. However, since there is a non-linear relationship between criterion position and hits and false alarms, this average will result in an underestimation of the hits and an overestimation of the false alarms. Combined, this results in a lower d' . **E)** Confidence increases as the criterion fluctuations become stronger (i.e., larger α and σ). Together with D, this suggests that criterion fluctuations can explain dissociation between accuracy and confidence.



Supplementary Figure 9: Criterion fluctuations map onto both drift bias and starting point. Linear mixed models were applied to each dataset separately. **A)** The main effect of drift bias was significant in all 14 datasets, indicating a strong and robust effect that generalizes across different experimental paradigms. **B)** In contrast, while the analysis across all datasets revealed a significant starting point bias, this effect was more difficult to detect at the level of individual datasets: only 4 out of 14 showed a

significant effect, and in one case the effect was in the opposite direction. **C)** Similarly, the interaction between drift bias and starting point bias reached significance in 5 out of 14 datasets, again with one effect in the opposite direction. The estimate plotted at the bottom, labelled “Total”, shows the parameter estimate from the model fitted to all datasets. Using nested linear mixed models regression we predicted the trial-to-trial criterion fluctuations based on the fluctuations in starting point and drift bias. When performing this analysis across all datasets, we see a significant effect of drift bias ($\beta = -0.308$, $SE = 0.002$, $F(1,412711) = 92231.85$, $p < .001$). Also the main effect for starting point bias is significant ($\beta = -0.358$, $SE = 0.001$, $F(1,412706) = 699.48$, $p < .001$). The negative slopes for both effects indicate that a positive drift bias or starting point, shifting evidence accumulation toward the upper boundary (“right” stimulus), corresponds to an SDT criterion shift to the left, thereby also making a “right” response more likely. Lastly, there is also a small but significant interaction between drift bias and starting point bias ($\beta = 0.025$, $SE = 0.001$, $F(1,412690) = 8.482$, $p < .004$), suggesting that if both drift bias and starting point bias steer the decision process towards the upper boundary (“right”), this results in a criterion fluctuating to the right, hereby decreasing the likelihood of a “right” response. Note: * = $p < .05$, ** = $p < .01$, *** = $p < .001$.



Supplementary Figure 10: Using the Hierarchical Model for Fluctuations in Criterion (hMFC) to recover criterion fluctuations with a linear trend over time. Although hMFC assumes an AR(1) process for the criterion fluctuations, the model can recover deviations from this assumption by virtue of the Forward Filtering/Backward Sampling algorithm. To demonstrate this, we simulated data from 50 subjects, each with 500 trials. The criterion fluctuations followed an AR(1) process with an additional linear trend over time. Despite this violation of the AR(1) stationarity assumption, the model successfully recovered the trended criterion trajectories, with strong correspondence between the true and inferred states (mean $r = .87$, $SE = .01$). The line of the inferred states indicates the posterior mean of each trial with the shaded area representing the 95% credible intervals.

Supplementary Tables

Supplementary Table 1: Overview of R-hat diagnostics per dataset to assess chain convergence of hMFC. All parameters for each dataset have a R-hat lower than 1.05, which suggests a proper mixing of the chains of the Gibbs sampler. Note that for μ_w and σ_w the highest R-hat is shown out of the five input variables.

Study	μ_w	σ_w	μ_a	σ_a	σ_{μ_x}
Adler & Ma (2018, experiment 1A)	1.001	1.000	1.035	1.026	1.000
Adler & Ma (2018, experiment 1B)	1.000	1.001	1.031	1.044	1.000
Adler & Ma (2018, experiment 3)	1.001	1.001	1.011	1.006	1.000
Calder-Travis et al. (unpublished)	1.002	1.004	1.008	1.018	1.000
Denison et al. (2018)	1.005	1.010	1.037	1.011	1.001
Filevich & Fandakova (unpublished)	1.001	1.001	1.008	1.008	1.000
Law & Lee (unpublished)	1.007	1.004	1.004	1.016	1.000
Maniscalco et al. (2017, experiment 2)	1.000	1.001	1.002	1.032	1.001
Orchard & Van Boxtel (2019, experiment 1)	1.008	1.024	1.020	1.008	1.001
Orchard & Van Boxtel (2019, experiment 2)	1.002	1.006	1.002	1.032	1.001
Recht et al. (unpublished)	1.003	1.001	1.047	1.009	1.000
Shekhar & Rahnev (2021, experiment 3)	1.003	1.001	1.006	1.039	1.000
Siedlecka et al. (2021)	1.001	1.000	1.001	1.005	1.000
Wang (2018)	1.002	1.001	1.019	1.000	1.001
Yeon et al. (unpublished, experiment 2)	1.007	1.011	1.003	1.002	1.000

Supplementary Table 2: Overview per study of the mean, standard deviation, minimum, and maximum for α , σ^2 , and μ_x . In hMFC the criterion fluctuations are assumed to follow a first-order autoregressive model. For each subject the autoregressive coefficient α , error variance σ^2 , and criterion trajectory mean μ_x is estimated. Note that μ_x is used for a closed-form estimation of the intercept b (for more technical details see Vloeberghs et al., 2025).

Study	α				σ^2				μ_x			
	Mean	Std	Min	Max	Mean	Std	Min	Max	Mean	Std	Min	Max
Adler & Ma (2018, experiment 1A)	0.82	0.11	0.63	0.96	0.11	0.07	0.05	0.30	-0.01	0.42	-0.70	0.66
Adler & Ma (2018, experiment 1B)	0.83	0.10	0.64	0.94	0.10	0.03	0.04	0.15	-0.05	0.44	-0.98	0.48
Adler & Ma (2018, experiment 3)	0.67	0.16	0.52	0.98	0.09	0.04	0.02	0.19	-0.21	0.39	-0.85	0.56
Calder-Travis et al. (unpublished)	0.87	0.02	0.84	0.95	0.14	0.11	0.05	0.49	-0.20	0.51	-1.83	1.01
Denison et al. (2018)	0.64	0.07	0.56	0.79	0.15	0.06	0.07	0.26	-0.07	0.40	-0.43	0.71
Filevich & Fandakova (unpublished)	0.61	0.08	0.52	0.89	0.13	0.04	0.09	0.23	-0.60	0.37	-1.54	-0.09
Law & Lee (unpublished)	0.82	0.08	0.70	0.95	0.69	0.74	0.11	2.53	-0.18	0.61	-1.41	0.86
Maniscalco et al. (2017, experiment 2)	0.91	0.03	0.85	0.96	0.12	0.07	0.04	0.43	0.09	0.52	-0.99	1.22
Orchard & Van Boxtel (2019, experiment 1)	0.85	0.08	0.71	0.98	0.29	0.42	0.03	2.37	0.02	0.46	-1.11	0.86
Orchard & Van Boxtel (2019, experiment 2)	0.89	0.06	0.72	0.97	0.34	0.88	0.04	5.04	0.13	0.80	-1.53	2.22
Recht et al. (unpublished)	0.74	0.04	0.70	0.80	0.21	0.17	0.06	0.56	0.03	0.15	-0.24	0.27
Shekhar & Rahnev (2021, experiment 3)	0.89	0.04	0.82	0.97	0.11	0.07	0.03	0.32	0.13	0.24	-0.34	0.53
Siedlecka et al. (2021)	0.84	0.08	0.68	0.98	0.08	0.03	0.04	0.19	0.02	0.29	-0.65	0.70
Wang (2018)	0.81	0.07	0.72	0.96	0.15	0.09	0.08	0.52	0.00	0.51	-1.22	0.68
Yeon et al. (unpublished, experiment 2)	0.88	0.02	0.84	0.92	0.46	0.96	0.05	5.47	0.23	0.43	-0.41	1.31

Supplementary Table 3: Investigating the robustness of the results for different values of the fixed hyperparameters μ_{σ^2} and β_{σ^2} in hMFC through a sensitivity analysis. We fitted hMFC to the data from Shekhar et al. (2021), while systematically varying the hyperparameter values ($\mu_{\sigma^2} = [0.05, 0.15, 0.25]$, $\beta_{\sigma^2} = [0.05, 0.15, 0.25]$). Note that the results reported in the paper are based on hMFC fits with $\mu_{\sigma^2} = 0.15$, $\beta_{\sigma^2} = 0.15$. The results indicate great robustness for different hyperparameter values, with little variability in the hMFC parameter estimates. For the parameter σ_a , the effective sample size (ESS) is notably lower compared to the other parameters. This is presumably due to a higher autocorrelation in the Gibbs sampler, as described in Supplementary Figure 6 in Vloeberghs et al. (2025). In the mixed models, the crucial interaction between stimulus direction and criterion is significant for all nine configurations, with nearly identical estimates. While not included in the table, also the other effects in the mixed models were identical. We therefore conclude that our results are robust for changes in the parameter values for μ_{σ^2} and β_{σ^2} .

		[μ_{σ^2} · β_{σ^2}]								
hMFC		[0.05, 0.05]	[0.05, 0.15]	[0.05, 0.25]	[0.15, 0.05]	[0.15, 0.15]	[0.15, 0.25]	[0.25, 0.05]	[0.25, 0.15]	[0.25, 0.25]
μ_w	Estimate	2.466	2.463	2.460	2.486	2.505	2.512	2.491	2.516	2.529
	R-hat	1.003	1.001	1.001	1.004	1.003	1.002	1.005	1.004	1.003
	ESS	13460.00	14970.00	15340.00	10230.00	13060.00	13810.00	8982.00	11520.00	12230.00
σ_w	Estimate	0.492	0.488	0.486	0.498	0.498	0.498	0.499	0.501	0.502
	R-hat	1.001	1.000	1.000	1.002	1.001	1.001	1.003	1.002	1.002
	ESS	11450.00	14130.00	14450.00	6554.00	9451.00	10860.00	5649.00	6249.00	6956.00
μ_a	Estimate	0.940	0.931	0.928	0.931	0.910	0.900	0.929	0.905	0.893
	R-hat	1.012	1.004	1.002	1.012	1.006	1.003	1.013	1.007	1.005
	ESS	382.30	629.60	764.00	378.40	598.10	638.10	367.8	614.80	674.40
σ_a	Estimate	0.062	0.049	0.042	0.073	0.069	0.065	0.075	0.076	0.076
	R-hat	1.038	1.026	1.015	1.066	1.041	1.034	1.071	1.066	1.036
	ESS	105.10	106.60	134.90	65.59	77.79	77.88	58.19	51.63	71.19
σ_{μ_x}	Estimate	0.406	0.406	0.405	0.401	0.408	0.409	0.410	0.410	0.410
	R-hat	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	ESS	15850.00	15920.00	15950.00	15860.00	15920.00	15950.00	15860.00	15920.00	15950.00
Mixed models										
Stimulus direction x criterion		-0.1371	-0.1358	-0.1354	-0.1366	-0.1342	-0.1331	-0.1364	-0.1338	-0.1324
p-value		< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

Supplementary Table 4: *Detailed description of empirical datasets.*

Study	Decision description	Stimulus strength manipulation	Additional manipulations	Number of subjects	Number of trials	Confidence rating type
Adler & Ma (2018, experiment 1A)	Distinguish the orientation of Gabors and ellipses drawn from two distributions with different means	Gabor contrast/ ellipse aspect ratio and orientation	-	11	2160	4-point discrete
Adler & Ma (2018, experiment 1B)	Distinguish the orientation of Gabors and ellipses drawn from two distributions from equal mean but distinct standard deviation	Gabor contrast/ ellipse aspect ratio and orientation	-	11	2160	4-point discrete
Adler & Ma (2018, experiment 3)	Distinguish the orientation of Gabors drawn from two distributions from equal mean but distinct standard deviation	Contrast and orientation	-	15	3240	4-point discrete
Calder-Travis et al. (unpublished)	Decide which of two groups of dots has more dots	Dot difference	Free response blocks: where participants could answer at any point during the trial, and forced response blocks: where a visual cue indicated when the participant had to respond	48	640	Continuous
Denison et al. (2018)	Distinguish the orientation of Gabors drawn from two distributions from equal mean but distinct standard deviation. In one trial, 4 Gabors were presented, and a response cue indicated which stimulus to decide on.	Contrast and orientation	Spatial cue that pointed to the Gabor that participants had to report (valid condition), to a different Gabor (invalid) or to all four possible locations (neutral condition).	12	1728-2160	4-point discrete

Filevich & Fandakova (unpublished)	2 screens with 6 gratings were presented, participants had to decide which of the two presentations contained one pop-out Gabor	Contrast (manipulated block wise)	-	17	600	6-point discrete
Law & Lee (unpublished)	Decide whether the dots are moving left or right	Motion coherence	-	16	1200	4-point discrete
Maniscalco et al. (2017, experiment 2)	Decide on which side of the screen a Gabor patch had appeared	Contrast	-	41	510	4-point discrete
Orchard & Van Boxtel (2019, experiment 1)	Decide whether the dots are moving left or right	Noisiness of dot motion	After 800 trials, all trials were repeated in the same order	45	1600	4-point discrete
Orchard & Van Boxtel (2019, experiment 2)	Decide whether the dots are moving left or right	Noisiness of dot motion	After 800 trials, all trials were repeated in the same order	37	1600	4-point discrete
Recht et al. (unpublished)	Decide on orientation of Gabor patch	Cue validity	-	10	1680	2-point discrete
Shekhar & Rahnev (2021, experiment 3)	Decide on orientation of Gabor patch	Contrast and orientation	-	20	2800	Continuous
Siedlecka et al. (2021)	Decide which of two groups of dots has more dots	Dot difference	The key needed to give a certain response was alternated every block	54	733-780	4-point discrete
Wang (2018)	Decide whether faces look fearful or happy	% of fear in the faces	The faces presented were either large faces or small faces, this was manipulated block wise.	24	504	3-point discrete
Yeon (unpublished, experiment 2)	Decide whether the dots are moving left or right	Volatility (manipulated block wise)	-	37	800-880	4-point discrete

